DATA SCIENTIST – MATRISK AI: PREDICTIVE MATERIAL INTELLIGENCE FOR FINANCIAL RISK MODELLING, COMMODITY VALUATION & INFRASTRUCTURE ASSET ANALYTICS

Category: Data Science | Material Sciences | Financial Risk Analytics | AI/ML

Timeline: 15 Days | AI-Accelerated Innovation & Research Project

Target: Undergraduate / MBA / Engineering Students (Python proficiency required)

TASK:

Build a dual-domain AI system that embeds material science intelligence - crystal structure encoding, physics-constrained property prediction, and material failure modelling - directly into financial risk models for commodity trading desks, infrastructure project finance, insurance catastrophe pricing, and ESG portfolio analytics. Design MatRisk Lab - a gamified simulation platform where learners manage a portfolio of infrastructure assets and commodity positions while responding to material-related financial shocks.

PART A: TRAINING MATERIAL & LEARNING RESOURCES

PART B: GAMIFIED SIMULATION BUSINESS CONCEPT

PART C: CASE STUDIES & REAL-WORLD APPLICATIONS

PART D: 15-DAY PROJECT METHODOLOGY & DELIVERABLES

PART E: NO-CODE TOOLS & PLATFORMS GUIDE

PART F: SUBMISSION PROTOCOL

Important Note:

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EXECUTIVE SUMMARY

This project positions students as Quantitative Material-Financial Analysts at a fictional cross-domain intelligence firm - MatRisk Capital Partners - requiring them to master the convergence of computational material science and quantitative financial analytics. Unlike traditional data science projects that operate within a single domain, MatRisk AI demands simultaneous fluency in crystallography, thermodynamics, and fracture mechanics on one side, and commodity derivatives pricing, credit risk modelling, and insurance loss estimation on the other.

The core challenge students must solve: How can financial institutions - commodity trading desks, infrastructure lenders, insurers, and ESG analysts - integrate the physical science governing the assets they trade, finance, and insure into their quantitative risk models? The answer lies in a purpose-built dual-domain AI architecture that processes material science databases (AFLOW, Materials Project, NIST) through Graph Neural Networks and Physics-Informed Neural Networks, then feeds the resulting intelligence into commodity price prediction, infrastructure credit risk scoring, insurance catastrophe modelling, and ESG impact assessment.

Students will build MatRisk Lab - a gamified simulation platform where learners manage a portfolio of infrastructure assets (bridges, pipelines, buildings) and commodity trading positions while responding to material-related financial shocks: supply crises, quality degradation events, climate-accelerated corrosion, and regulatory material mandates. An AI opponent using the MatRisk AI models competes against the player, demonstrating the value of material science integration in financial decision-making.

The global material science market is valued at approximately $6.2 trillion, while the financial instruments that trade, insure, and finance material-dependent assets represent an estimated $40+ trillion in notional value. Yet these two domains operate in near-complete isolation. A steel futures trader prices Hot Rolled Coil contracts without understanding how microstructural phase transitions affect supply dynamics. An insurer underwrites a bridge portfolio without modelling fatigue-crack propagation physics. MatRisk AI closes this intelligence gap - creating a first-of-its-kind convergence system with multiple patentable innovations.

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PART A: TRAINING MATERIAL & LEARNING RESOURCES

SECTION A1: DOMAIN FUNDAMENTALS - THE MATERIAL SCIENCE FOUNDATION

A1.1 Introduction To Computational Material Science

Computational material science is the discipline of using mathematical models, physics simulations, and machine learning to predict, discover, and optimise materials without requiring expensive physical experimentation. In the context of MatRisk AI, the material science foundation provides the physical intelligence layer that financial models currently lack entirely.

Definition And Core Concept:

Material science studies the relationship between a material's composition (what atoms are present and in what proportions), its structure (how those atoms are arranged - crystal structure, grain boundaries, defects), its processing (how it was manufactured - temperature, pressure, cooling rate), and its properties (what it can do - strength, conductivity, corrosion resistance). This four-way relationship - known as the Materials Tetrahedron - is the foundational framework for all material analysis.

Key Characteristics Of Material Data:

- Materials are described by their crystal structure - the repeating 3D arrangement of atoms in a lattice, classified into 7 crystal systems and 230 space groups
- Each material has a unique composition expressed as a chemical formula
- Properties are not independent - they are governed by physics: bulk modulus and shear modulus are related through Poisson's ratio; formation energy determines thermodynamic stability
- Material behaviour is multi-scale: atomic-level bonding determines bulk mechanical properties; microstructural grain boundaries determine fatigue life and fracture behaviour
- Unlike financial data which is time-series-based, material data is primarily compositional and structural - requiring specialised encoding techniques (graph representations, atomic descriptors)
- Material databases contain millions of entries computed via Density Functional Theory (DFT) - quantum mechanical calculations that predict properties from first principles

The Materials Tetrahedron:

Composition ↔ Structure ↔ Processing ↔ Properties

For MatRisk AI, we primarily work along the Composition → Structure → Properties axis, using machine learning to predict properties from composition and structure without requiring costly DFT calculations or physical experiments. The Processing dimension is incorporated through manufacturing condition features where available.

A1.2 Crystal Structures And Lattice Systems

Every crystalline material has atoms arranged in a periodic, repeating pattern called a crystal lattice. The lattice is characterised by its unit cell - the smallest repeating unit that, when tiled in three dimensions, reproduces the entire crystal. Understanding crystal structure is essential because it directly determines a material's mechanical, electronic, thermal, and optical properties.

The 7 Crystal Systems:

- **Cubic:** $a = b = c$, $\alpha = \beta = \gamma = 90°$. Most symmetric. Includes FCC (face-centred cubic - aluminium, copper, gold), BCC (body-centred cubic - iron, tungsten), and diamond cubic (silicon, germanium). FCC metals are typically ductile; BCC metals are strong but less ductile.
- **Hexagonal:** $a = b \neq c$, $\alpha = \beta = 90°$, $\gamma = 120°$. Includes HCP (hexagonal close-packed - titanium, magnesium, zinc). HCP metals often exhibit anisotropic mechanical behaviour - different strength in different directions.
- **Tetragonal:** $a = b \neq c$, $\alpha = \beta = \gamma = 90°$. Stretched cubic. Includes tin (white) and many piezoelectric ceramics ($BaTiO_3$).
- **Orthorhombic:** $a \neq b \neq c$, $\alpha = \beta = \gamma = 90°$. Three unequal axes at right angles. Includes sulphur and many minerals.
- **Monoclinic:** $a \neq b \neq c$, $\alpha = \gamma = 90°$, $\beta \neq 90°$. One oblique angle. Includes gypsum.
- **Triclinic:** $a \neq b \neq c$, $\alpha \neq \beta \neq \gamma$. Least symmetric. Includes many feldspars.
- **Trigonal (Rhombohedral):** $a = b = c$, $\alpha = \beta = \gamma \neq 90°$. Includes quartz, calcite, and bismuth.

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Why Crystal Structure Matters For Financial Analysts:

Crystal structure determines the commodity grade and quality premiums that drive pricing in metals markets. FCC metals (aluminium, copper) command different price dynamics than BCC metals (steel, tungsten) because their mechanical properties - ductility, work-hardening behaviour, fatigue resistance - differ fundamentally. When a steel mill shifts production from one phase (austenitic/FCC) to another (ferritic/BCC), the downstream quality impact cascades into futures pricing. MatRisk AI encodes this physics directly into financial signal generation.

A1.3 Material Properties And Their Physical Constraints

A fund accountant must know NAV computation rules; a material-financial analyst must know property definitions and their physical bounds. The following properties are the primary prediction targets for MatRisk AI's models:

Understanding The Property Hierarchy:

Material properties exist in a hierarchy of dependency. Atomic-level properties (electronegativity, atomic radius, ionisation energy) determine bonding character. Bonding determines crystal structure and electronic band structure. Structure determines bulk mechanical properties (strength, hardness, ductility). Mechanical properties determine engineering performance (fatigue life, creep resistance, corrosion behaviour). Engineering performance determines financial value (infrastructure RUL, commodity grade premium, insurance risk rating). MatRisk AI's multi-layer architecture mirrors this hierarchy: the GNN learns atomic and structural patterns; the PINN enforces physical relationships between properties; the financial modules translate properties into financial metrics.

The Elastic Moduli Relationships (Critical For Pinn):

The three elastic constants - Young's Modulus (E), Bulk Modulus (K), and Shear Modulus (G) - are not independent for isotropic materials. They are related through Poisson's Ratio (ν):

$$E = 2G(1 + \nu) = 3K(1 - 2\nu)$$

$$G = E / (2(1 + \nu)) = 3K(1 - 2\nu) / (2(1 + \nu))$$

$$K = E / (3(1 - 2\nu)) = 2G(1 + \nu) / (3(1 - 2\nu))$$

These relationships mean that if our model predicts K = 150 GPa, G = 80 GPa, and ν = 0.30, we can verify consistency: E from K = 3 × 150 × (1 − 0.60) = 180 GPa. E from G = 2 × 80 × 1.30 = 208 GPa. These differ by 15%, indicating an inconsistency that the PINN must resolve. The physics loss penalises this inconsistency, forcing the model to learn the true relationships rather than predicting each property independently.

Alloy Design Principles For The GAN:

The GAN's inverse design capability is grounded in established metallurgical principles:

- **Solid Solution Strengthening:** Adding atoms of different size to a base metal creates lattice distortion that impedes dislocation motion, increasing strength. The magnitude depends on the atomic size mismatch: larger mismatch = more strengthening. Example: adding 6% Al to Ti (atomic radius 1.76Å vs 1.43Å for Al) creates significant lattice strain.
- **Precipitation Hardening:** Creating nanoscale precipitate particles within the metal matrix that block dislocation movement. Requires specific alloy compositions that form coherent precipitates during aging heat treatment. Example: Ni_3Al (gamma-prime) precipitates in nickel superalloys provide strength retention up to 1000°C.
- **Grain Boundary Engineering:** Controlling the size and character of grain boundaries to optimise the trade-off between strength (smaller grains = stronger, Hall-Petch relationship) and ductility (larger grains = more ductile). The Hall-Petch equation: $\sigma_y = \sigma_0 + k_y / \sqrt{d}$, where d is grain size.
- **Phase Engineering:** Controlling which crystal phases are present (e.g., austenite vs. ferrite vs. martensite in steel) through composition and heat treatment. Different phases have radically different properties: austenite is soft and ductile; martensite is hard and brittle. The GAN can learn which compositions stabilise desired phase mixtures.

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Commodity-Material Property Linkages:

The following table maps specific material properties to their financial market implications - the conceptual foundation for MatRisk AI's cross-domain signal generation:

Material Property	Change Direction	Commodity Impact	Financial Signal Type
Steel Tensile Strength	Decrease (quality degradation)	HRC quality premium compression	Short signal on steel futures quality spread
Aluminium Corrosion Rate	Increase (environmental stress)	Higher maintenance costs for Al structures	Increased insurance premiums, reduced infrastructure RUL
Rare Earth Magnetic Moment	Decrease (substitute material)	Rare earth demand reduction	Short rare earth, long substitute material
Copper Conductivity	Stable (no change)	No signal	Null signal (no trade)
Battery Cathode Energy Density	Increase (new chemistry)	Lithium demand shift if cobalt-free	Short cobalt, long lithium/iron phosphate
Concrete Compressive Strength	Decrease (aging/carbonation)	Building failure risk increase	Increased insurance reserves, credit risk escalation
Titanium Fatigue Life	Decrease (microstructural change)	Aerospace maintenance acceleration	Short airline equities, long titanium substitutes

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Mechanical Properties:

- **Bulk Modulus (K):** Resistance to uniform compression. Unit: GPa. Physical constraint: $K > 0$ (thermodynamic stability requirement). Range: 1–450 GPa. Diamond has the highest bulk modulus (~440 GPa); rubber is near 1 GPa. Financial relevance: determines material suitability for pressure-bearing applications (pipelines, pressure vessels) that constitute large infrastructure loan portfolios.

- **Shear Modulus (G):** Resistance to shape change under shear stress. Unit: GPa. Physical constraint: $G > 0$ (mechanical stability). Range: 0.5–250 GPa. Financial relevance: determines structural integrity under lateral loads (earthquake resistance, wind loading) - critical for insurance catastrophe modelling.

- **Poisson's Ratio (ν):** Ratio of transverse strain to axial strain under uniaxial stress. Dimensionless. Physical constraint: $-1 < \nu < 0.5$ (thermodynamic bounds from Cauchy stress tensor symmetry). Most engineering materials: 0.2–0.35. Auxetic materials: $\nu < 0$. The elastic moduli are related: $E = 2G(1 + \nu) = 3K(1 - 2\nu)$. This relationship is enforced as a physics constraint in the PINN.

- **Tensile Strength (σ_{UTS}):** Maximum stress a material can withstand before fracture. Unit: MPa. Range: 10–3000 MPa. Must exceed yield strength. Financial relevance: directly determines structural component ratings used in infrastructure credit risk scoring.

- **Yield Strength (σ_y):** Stress at which permanent deformation begins. Unit: MPa. Range: 5–2500 MPa. Financial relevance: the design threshold for structural components; exceedance triggers failure progression.

Electronic And Thermodynamic Properties:

- **Band Gap (E_g):** Energy gap between valence and conduction bands. Unit: eV. Metals: $E_g = 0$; Semiconductors: $0 < E_g < 4$ eV; Insulators: $E_g > 4$ eV. Range: 0–12 eV. Financial relevance: determines material application in semiconductor and energy sectors - affecting technology commodity demand forecasts.

- **Formation Energy (E_f):** Energy change when a compound forms from its constituent elements. Unit: eV/atom. Stable compounds: $E_f < 0$. Range: -5.0 to +2.0 eV/atom. Financial relevance: determines thermodynamic stability of alloys - materials with marginally negative formation energy are susceptible to decomposition under stress, creating insurance-relevant failure modes.

- **Thermal Conductivity (κ):** Rate of heat transfer through a material. Unit: W/(m·K). Physical constraint: $\kappa > 0$ (second law of thermodynamics). Range: 0.01–430 W/(m·K). Financial relevance: determines material suitability for thermal applications (power plant components, heat exchangers) and fire resistance ratings used in insurance underwriting.

Fatigue And Failure Properties:

- **Fatigue Life:** Number of stress cycles before failure under cyclic loading. Governed by Paris' Law: da/dN = C(ΔK)^m, where da/dN is crack growth rate per cycle, ΔK is stress intensity factor range, and C, m are material constants. Financial relevance: directly determines remaining useful life (RUL) of infrastructure components - the primary input to the credit risk survival model.
- **Corrosion Rate:** Rate of material degradation in a given environment. Unit: mm/year. Depends on material composition, environmental conditions (temperature, humidity, chemical exposure), and protective coatings. Financial relevance: primary driver of infrastructure maintenance costs and RUL estimation for coastal and industrial-zone structures.
- **Creep Rate:** Rate of slow, permanent deformation under sustained stress at elevated temperatures. Financial relevance: critical for power plant and aerospace component life estimation.

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A1.4 Material Databases And Data Ecosystem

The material science community has built several large-scale computational databases that serve as the data foundation for MatRisk AI. These databases contain properties computed via Density Functional Theory (DFT) - quantum mechanical simulations that predict material properties from first principles - as well as experimentally measured values.

Primary Databases:

- **AFLOW (Automatic FLOW):** The largest computational materials database with over 3.5 million compound entries. Provides formation energy, band gap, bulk modulus, magnetic properties, and electronic structure data. Accessed via REST API at aflow.org. Key advantage: breadth of coverage across the periodic table and compound types.
- **Materials Project:** 150,000+ materials with crystal structure, electronic, thermodynamic, and mechanical properties. Developed at Lawrence Berkeley National Laboratory. Accessed via Python API client (mp-api). Key advantage: high-quality curated data with standardised DFT calculation parameters and extensive documentation.
- **NIST Standard Reference Data:** Multiple curated datasets covering thermophysical properties, phase diagrams, corrosion data, and mechanical testing results. Key advantage: experimentally validated data (not DFT-computed), providing ground truth for model validation.
- **Citrine Informatics:** 300,000+ records covering mechanical properties, processing conditions, and alloy compositions. Key advantage: includes processing parameters that other databases lack, enabling the full Materials Tetrahedron analysis.
- **JARVIS-DFT (NIST):** 75,000+ materials including specialised datasets for 2D materials, solar cell materials, and topological materials. Key advantage: cutting-edge material categories relevant to emerging technology sectors.

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SECTION A2: FINANCIAL ANALYTICS FOUNDATION

A2.1 Commodity Derivatives And Material Markets

Commodity trading is the financial activity most directly connected to material science - yet commodity traders rarely incorporate material science intelligence into their models. MatRisk AI bridges this gap by generating material-science-derived alternative data signals that enhance commodity price prediction.

Key Commodity Markets Relevant To Material Science:

- **Steel (HRC Futures):** Hot Rolled Coil is the benchmark steel product traded on CME, LME, and SHFE. Price is driven by iron ore cost, coal (coking) cost, production capacity utilisation, and quality premiums based on steel grade (carbon content, alloy additions). MatRisk AI's material science signals track production quality shifts that precede price movements.
- **Aluminium (LME):** Primary aluminium traded on London Metal Exchange. Price driven by energy costs (smelting is energy-intensive), bauxite supply, and demand from aerospace/automotive sectors. Material science signal: alloy recycling rates and recycled-vs-virgin quality differentials affect supply quality and ESG premiums.
- **Copper (COMEX):** Primary copper traded on COMEX. Essential for electrical applications due to high conductivity. Material science signal: substitution threat from aluminium alloys (when engineered to sufficient conductivity) creates cross-commodity price dynamics.
- **Rare Earth Elements:** Neodymium, dysprosium, lanthanum - critical for permanent magnets, EV motors, and wind turbines. Extreme supply concentration (China controls ~70% of production). Material science signal: alternative material discoveries (iron nitride magnets, ferrite substitutes) can cause structural demand shifts.
- **Lithium & Cobalt:** Battery materials for EVs and energy storage. Material science signal: solid-state battery research and sodium-ion alternatives threaten lithium demand; cobalt-free cathode designs (LFP, LNMO) already shifting cobalt demand curves.

A2.2 Credit Risk Modelling For Infrastructure

Infrastructure credit risk - the probability that a bridge, pipeline, or building will require significant repair or replacement, causing loan default - is fundamentally a material degradation problem. Yet current credit risk models use statistical approaches (historical default rates, regression-based scorecards) that ignore the physics of material failure entirely.

The Credit Risk Framework:

Expected Loss (EL) = Probability of Default (PD) × Loss Given Default (LGD) × Exposure at Default (EAD)

Probability Of Default (PD):

In traditional banking, PD is estimated from historical default frequencies and borrower characteristics (financial ratios, industry, credit scores). For infrastructure loans, MatRisk AI replaces this with a physics-informed PD derived from the survival model: $PD(t) = 1 - S(t|x)$, where $S(t|x)$ is the survival function computed from material properties, environmental conditions, and operational loads.

Loss Given Default (LGD):

The fraction of exposure lost if failure occurs. For infrastructure, LGD is modelled as a function of failure severity - which depends on the material failure mode. A ductile failure (gradual deformation before fracture) has lower LGD than a brittle failure (sudden catastrophic fracture with no warning). MatRisk AI predicts failure mode from material properties, enabling more accurate LGD estimation.

Exposure At Default (EAD):

Outstanding loan balance at the time of failure. Computed from the amortisation schedule. For revolving credit facilities (common in infrastructure project finance), EAD includes unfunded commitments likely to be drawn.

A2.3 Insurance And Catastrophe Modelling

Insurance catastrophe modelling estimates the probability and severity of large-scale loss events. For engineering and infrastructure insurance (covering bridges, buildings, pipelines, industrial plants), the insured event is typically structural failure caused by material degradation, overloading, environmental attack, or a combination.

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Current Approach (Actuarial):

Insurers currently estimate structural failure probabilities using historical claims data and actuarial tables. This approach has fundamental limitations: historical data is sparse (catastrophic failures are rare events), it cannot account for material-specific degradation patterns, and it treats all structures of a given type as statistically identical regardless of their actual material condition.

Matrisk AI Approach (Physics-Informed):

MatRisk AI replaces actuarial averages with physics-informed failure probability distributions. For each insured structure, the system predicts a full survival curve S(t|x) based on the structure's material properties, age, environmental exposure, and operational loads. The insurance pricing module then computes:

- **Loss Frequency:** The expected number of failure events per year = 1 − S(1|x) for annual policies.
- **Loss Severity Distribution:** Log-normal distribution parameterised by material type, structure age, and failure mode. Ductile failures produce lower mean severity; brittle failures produce higher mean and fatter tails.
- **Aggregate Loss:** Compound Poisson distribution - frequency × severity - giving the total expected loss across a portfolio of insured structures.
- **Tail Value at Risk (TVaR):** Expected loss in the tail beyond the 99.5th percentile - the key metric for insurance regulators and reserve estimation.

A2.4 ESG And Sustainability Analytics

Environmental, Social, and Governance (ESG) analysis increasingly influences capital allocation, with over $35 trillion in ESG-mandated assets globally. Material substitution - replacing one material with another that has lower environmental impact - is a primary lever for reducing portfolio carbon intensity. Yet current ESG analytics treat material choices as binary (e.g., "uses recycled materials: yes/no") without quantifying the property trade-offs and cost implications.

Matrisk AI ESG Module:

- **Carbon Intensity Calculator:** CO_2-equivalent emissions per kilogram for each material, sourced from lifecycle assessment databases. Virgin aluminium: ~12 kg CO_2/kg; recycled aluminium: ~0.6 kg CO_2/kg. Steel (BOF): ~2.0 kg CO_2/kg; Steel (EAF/recycled): ~0.4 kg CO_2/kg.
- **Substitution Impact Analyser:** Given a material switch (e.g., virgin → recycled aluminium), compute: (a) carbon intensity reduction, (b) material property impact using CGNN predictions, (c) cost impact from commodity price differentials, (d) portfolio-level ESG score change.
- **Green Bond Eligibility Scorer:** Assess infrastructure projects against ICMA Green Bond Principles based on their material choices, energy efficiency of material production, and lifecycle environmental impact.

SECTION A3: MACHINE LEARNING ARCHITECTURE FOR MATERIAL-FINANCIAL ANALYTICS

A3.1 Graph Neural Networks For Crystal Structure Encoding

Graph Neural Networks (GNNs) are the state-of-the-art approach for encoding crystal structures because they naturally represent the non-Euclidean, variable-size, permutation-invariant nature of atomic arrangements. In a crystal graph, atoms are nodes and bonds are edges, allowing the network to learn spatial relationships that govern material properties.

Crystal Graph Construction:

- **Nodes (Atoms):** Each atom in the unit cell becomes a node with a feature vector encoding: atomic number, electronegativity (Pauling scale), atomic radius, ionisation energy, electron affinity, group number, period number, and block encoding (s, p, d, f). Total node feature dimension: 8–12 features per atom.
- **Edges (Bonds):** Bonds are defined between atoms within a cutoff radius (typically 8Å). Edge features encode: interatomic distance, bond type (ionic, covalent, metallic), and Gaussian expansion of the distance (converting a single scalar into a multi-dimensional feature using evenly spaced Gaussian basis functions). Total edge feature dimension: 40–80 features per bond.
- **Global State:** Composition-level descriptors (Magpie features) appended as a global context vector, providing the network with information about the overall composition that individual atom-bond message passing might miss.

Message Passing Mechanism:

The GNN updates node representations through iterative message passing: each atom "receives messages" from its bonded neighbours, aggregates them, and updates its own representation. After L rounds of message passing, each atom's representation encodes information about its L-hop neighbourhood in the crystal. A global pooling operation (attention-weighted mean) aggregates all atom representations into a single crystal-level representation that feeds into the prediction heads.

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$$h_i^{(l+1)} = \sigma(W_{self} \cdot h_i^{(l)} + \sum_{j \in N(i)} W_{edge} \cdot [h_j^{(l)} \| e_{ij}])$$

Where h_i is the representation of atom i, $N(i)$ is its set of neighbours, e_{ij} is the edge feature between atoms i and j, W_{self} and W_{edge} are learnable weight matrices, and σ is a non-linear activation function.

A3.2 Physics-Informed Neural Networks (PINNs)

Physics-Informed Neural Networks embed known physical laws directly into the neural network's training objective as soft constraints. This ensures that model predictions are not just statistically accurate but also physically plausible - a critical requirement for financial applications where physically impossible predictions would undermine credibility with domain experts.

Physics Constraints Implemented In Matrisk AI:

- **Elastic Moduli Consistency:** $E = 2G(1 + \nu) = 3K(1 - 2\nu)$. If the model predicts bulk modulus K, shear modulus G, and Poisson's ratio ν, these three values must satisfy this thermodynamic relationship. The physics loss penalises deviations: L_elastic = |2G(1+ν) − 3K(1−2ν)|^2.
- **Thermodynamic Consistency:** Gibbs free energy $G = H - TS$. Predictions of enthalpy H, entropy S, and Gibbs energy G must satisfy this fundamental thermodynamic identity. L_thermo = |G_pred − (H_pred − T·S_pred)|^2.
- **Poisson's Ratio Bounds:** $-1 < \nu < 0.5$. Thermodynamic stability requires Poisson's ratio to lie strictly within these bounds. The loss function includes a penalty for predictions outside this range: L_bounds = max(0, ν − 0.5)^2 + max(0, −1 − ν)^2.
- **Paris' Law for Fatigue:** $da/dN = C(\Delta K)^m$. Crack growth rate predictions must follow this established fracture mechanics relationship, with material-specific constants C and m. The PINN learns to predict C and m from composition/structure.

Total PINN Loss Function:

$$\mathbf{L_total = L_data + \lambda_1 L_elastic + \lambda_2 L_thermo + \lambda_3 L_bounds + \lambda_4 L_fatigue}$$

The λ weights are dynamically adjusted using the GradNorm algorithm to balance learning across data-fitting and physics-constraint objectives. Early in training, data loss dominates; as the model improves, physics losses receive increasing weight to refine predictions within physically plausible boundaries.

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A3.3 Generative Adversarial Networks For Inverse Material Design

The GAN module addresses the inverse design problem: given a target property profile (e.g., tensile strength > 1200 MPa, density < 4.5 g/cm³, cost < $15/kg), generate candidate alloy compositions that satisfy all constraints simultaneously.

Architecture - Wasserstein Gan With Gradient Penalty (WGAN-GP):

- **Generator:** Takes a noise vector z ~ N(0,1) concatenated with a target property vector and a cost budget scalar. Outputs a composition vector (fractions of each element) through a Softmax layer ensuring compositions sum to 1. Architecture: 4-layer MLP with LayerNorm and LeakyReLU.
- **Critic (Discriminator):** Takes a composition vector and outputs a scalar quality score. Architecture: 3-layer MLP with spectral normalisation. Trained to distinguish real compositions (from databases) from generated ones.
- **Property Predictor:** A frozen copy of the trained CGNN that provides gradient signal back to the generator, ensuring generated compositions actually achieve target properties.

Financial Cost Constraint:

The GAN's loss function includes a cost term using real-time commodity prices for each constituent element. The cost of a generated alloy is computed as: Cost = ∑(composition_fraction_i × element_price_i). A penalty is applied when Cost exceeds the target budget. This creates a direct, novel bridge between material design and financial markets - generated alloys are not just physically optimal but economically viable at current market prices.

A3.4 Survival Analysis For Infrastructure Remaining Useful Life

Survival analysis models the time until an event occurs - in MatRisk AI, the time until a structural component fails. The Deep Survival Network combines the interpretability of Cox Proportional Hazards with the nonlinear capacity of deep learning to produce full survival curves for each infrastructure component.

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Cox Proportional Hazards Model:

$$\mathbf{h(t|x) = h_0(t) \cdot \exp(\beta' x)}$$

Where h(t|x) is the hazard function (instantaneous failure rate), $h_0(t)$ is the baseline hazard, and β'x is the linear predictor from covariates x. The survival function is: $S(t|x) = \exp(-H_0(t) \cdot \exp(\beta' x))$, where $H_0(t)$ is the cumulative baseline hazard.

Deep Survival Extension:

The DeepSurv model replaces the linear predictor β'x with a neural network f(x), enabling nonlinear covariate interactions: $h(t|x) = h_0(t) \cdot \exp(f(x))$. The neural network takes as input: material property predictions from the CGNN, environmental condition features (climate zone, chemical exposure), structural design parameters (load capacity, safety margins), and operational data (traffic volume, maintenance history).

Financial Integration:

The survival curves are directly converted into financial credit risk metrics. For a bridge with survival function S(t|x): PD at year 5 = 1 − S(5|x). For an infrastructure loan portfolio, the portfolio-level PD is computed by aggregating individual component survival curves weighted by exposure. This enables the first-ever physics-informed infrastructure credit risk scoring system.

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SECTION A4: DATA SOURCES AND ACQUISITION STRATEGY

A4.1 Material Science Data Sources

MatRisk AI requires data from both material science and financial domains. The material science data provides the physical intelligence foundation; the financial data provides the market context for translating material properties into financial signals.

Database	Entries	Key Properties	Access Method
AFLOW	3.5M+ compounds	Formation energy, band gap, bulk modulus, magnetic properties	REST API (aflow.org)
Materials Project	150K+ materials	Crystal structure, electronic, thermodynamic, mechanical	MP API (Python client)
NIST SRD	Multiple datasets	Thermophysical, phase diagrams, corrosion	Web + bulk download
Citrine Informatics	300K+ records	Mechanical, processing conditions, alloys	API (academic license)
JARVIS-DFT	75K+ materials	2D materials, solar cells, topological	REST API + bulk download
Open Quantum Materials DB	1M+ calculations	DFT-computed properties, defect energies	REST API
National Bridge Inventory	620K+ US bridges	Material type, condition ratings, traffic	Public CSV download

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A4.2 Financial Data Sources

Source	Coverage	Key Fields	Access
Yahoo Finance / yfinance	Global commodity prices	OHLCV, futures, options	Free API
Quandl (Nasdaq Data Link)	Metals, energy, agriculture	Spot, forwards, inventory	Free + premium tiers
FHWA Infrastructure Data	US road/bridge network	Structural deficiency, repair history	Public download
Swiss Re / Munich Re Cat	Global catastrophes	Loss amounts, cause, location	Published reports
World Bank ESG Data	180+ countries	Carbon intensity, sustainability metrics	Open data API
USGS Mineral Commodity	90+ minerals	Production, consumption, price, reserves	Public download

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A4.3 Data Quality Framework

All data must pass through a rigorous quality gate before entering the model training pipeline. The data quality framework enforces:

- **Completeness:** Minimum 80% feature completeness required per entry. Entries below threshold are flagged for review, not silently dropped.
- **Physical Plausibility:** Automated bounds checking against known physical constraints (e.g., negative formation energy for stable compounds, Poisson's ratio between -1 and 0.5, density > 0). Violations trigger immediate investigation.
- **Outlier Detection:** Isolation Forest combined with domain-informed rules. Outliers are investigated, not automatically removed - they may represent genuine novel materials.
- **Cross-Source Validation:** Properties reported by multiple databases are cross-validated. Discrepancies > 10% are flagged for manual review.
- **Temporal Alignment:** Financial data timestamps aligned to material database publication dates to prevent look-ahead bias in backtesting.
- **Version Control:** All datasets version-controlled via DVC with SHA-256 hashes for full reproducibility.

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SECTION A4-DS: SAMPLE DATASETS AND DATA DICTIONARIES

This section provides the sample datasets that interns will use for development, testing, and simulation. These datasets are either pre-built fictional datasets (for the gamified simulator) or documented extraction templates (for real API sources). Interns must use these exact schemas and extend them with real data from the APIs listed in Section A4.

DS-1: Sample Material Properties Dataset (50 Entries)

The following table provides a representative sample of the material property data that the CGNN and ensemble models will be trained on. This sample covers common structural metals, ceramics, semiconductors, and alloys. Interns should verify these values against Materials Project and use this as a validation reference.

Material	Formula	Crystal System	Space Group	E_f (eV/atom)	E_g (eV)	K (GPa)	G (GPa)	ν	Density (g/cm³)
Iron (BCC)	Fe	Cubic	Im-3m (229)	-0.00	0.00	170	82	0.29	7.87
Aluminium	Al	Cubic	Fm-3m (225)	-0.00	0.00	76	26	0.35	2.70
Copper	Cu	Cubic	Fm-3m (225)	-0.00	0.00	140	48	0.34	8.96
Titanium	Ti	Hexagonal	$P6_3/mmc$ (194)	-0.00	0.00	110	44	0.32	4.51
Silicon	Si	Cubic	Fd-3m (227)	-0.46	1.12	98	66	0.22	2.33
Nickel	Ni	Cubic	Fm-3m (225)	-0.00	0.00	180	76	0.31	8.91
Tungsten	W	Cubic	Im-3m (229)	-0.00	0.00	310	161	0.28	19.25
Hematite	Fe_2O_3	Trigonal	R-3c (167)	-2.83	2.20	178	90	0.27	5.26

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Corundum	Al_2O_3	Trigonal	R-3c (167)	-3.47	8.80	254	163	0.24	3.99
Titanium Dioxide	TiO_2	Tetragonal	$P4_2/mnm$ (136)	-3.46	3.05	220	110	0.28	4.23
Sodium Chloride	NaCl	Cubic	Fm-3m (225)	-3.43	5.00	25	13	0.25	2.16
Gallium Arsenide	GaAs	Cubic	F-43m (216)	-0.65	1.42	75	33	0.31	5.32
Zinc Oxide	ZnO	Hexagonal	$P6_3mc$ (186)	-3.10	3.37	143	44	0.36	5.61
Calcium Fluoride	CaF_2	Cubic	Fm-3m (225)	-6.10	11.80	83	42	0.27	3.18
Magnesium Oxide	MgO	Cubic	Fm-3m (225)	-5.99	7.80	160	131	0.18	3.58

Data Dictionary - Material Properties:

- **material_id:** Unique identifier from Materials Project (format: mp-XXXXX) or AFLOW (format: aflow:XXXXXXXX)
- **formula_pretty:** Reduced chemical formula (e.g., Fe_2O_3 not Fe4O6)
- **crystal_system:** One of: cubic, hexagonal, tetragonal, orthorhombic, monoclinic, triclinic, trigonal
- **spacegroup_number:** Integer 1–230 identifying the space group symmetry
- **formation_energy_per_atom:** eV/atom. Negative = stable. Computed via DFT. Source: Materials Project.
- **band_gap:** eV. 0 = metal. >0 = semiconductor/insulator. Source: Materials Project (PBE functional, known to underestimate).
- **bulk_modulus_vrh:** GPa. Voigt-Reuss-Hill average. Source: Materials Project elasticity database.

- **shear_modulus_vrh:** GPa. Voigt-Reuss-Hill average. Must satisfy: $G > 0$ and $E = 2G(1+\nu)$.
- **poisson_ratio:** Dimensionless. Must satisfy: $-1 < \nu < 0.5$. Computed from K and G.
- **density:** g/cm³. Computed from unit cell volume and composition. Must be > 0.

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DS-2: Sample Commodity Price Dataset (30-Day Window)

The following table provides a representative sample of daily commodity price data for Steel HRC futures. Interns will download 10+ years of similar data via yfinance. This sample illustrates the format and typical daily movements.

Date	HRC Close ($/ton)	Daily Return	5D Vol (%ann)	RSI(14)	Bollinger Z	Term Spread ($/ton)
2024-01-02	985.50	+0.32%	22.4%	55.2	+0.35	-12.50
2024-01-03	991.00	+0.56%	23.1%	57.8	+0.62	-11.00
2024-01-04	978.25	-1.29%	28.5%	48.1	-0.45	-14.75
2024-01-05	982.00	+0.38%	27.8%	49.5	-0.22	-13.50
2024-01-08	995.50	+1.37%	30.2%	58.3	+0.78	-8.25
2024-01-09	1002.75	+0.73%	29.5%	62.1	+1.15	-5.50
2024-01-10	998.00	-0.47%	26.3%	59.4	+0.87	-7.00
2024-01-11	1005.25	+0.73%	25.1%	63.7	+1.32	-4.25
2024-01-12	1010.50	+0.52%	23.8%	66.2	+1.58	-2.75
2024-01-16	1003.00	-0.74%	24.5%	60.5	+1.02	-5.50

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Data Dictionary - Commodity Prices:

- **date:** Trading date (business days only). Format: YYYY-MM-DD.
- **close:** Daily closing price in USD per metric ton. Source: CME HRC futures via yfinance (ticker: HRC=F).
- **daily_return:** Log return = ln(close_t / close_{t-1}). First day of series is NaN.
- **vol_5d:** 5-day rolling annualised volatility = std(daily returns over 5 days) × √252 × 100.
- **rsi_14:** 14-day Relative Strength Index. >70 = overbought, <30 = oversold.
- **bollinger_z:** Z-score relative to 63-day SMA and standard deviation. >2 = price extended above average.
- **term_spread:** Second month futures − front month futures. Negative = backwardation (supply tight). Positive = contango.

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DS-3: Sample Infrastructure Portfolio Dataset (10 Assets)

The following dataset provides the 10 fictional infrastructure assets used in the MatRisk Lab gamified simulator. Each asset has material composition, environmental exposure, and financial characteristics. Interns must use these exact assets for the simulator and extend with real NBI data for the credit risk model.

Asset ID	Type	Material	Age (yr)	Location	Condition (1–9)	Loan ($M)	Annual Traffic
BR-001	Highway Bridge	A36 Structural Steel	42	Coastal Florida	5	12.0	35,000 ADT
BR-002	Rail Bridge	Weathering Steel A588	28	Industrial Ohio	6	8.5	45 trains/day
BR-003	Pedestrian Bridge	Stainless Steel 316L	15	Urban Chicago	7	3.2	5,000 ped/day
PL-001	Oil Pipeline	API 5L Grade X65	35	Coastal Texas	4	45.0	250K bbl/day
PL-002	Gas Pipeline	API 5L Grade X70	22	Rural Wyoming	7	28.0	1.2 BCF/day
BL-001	Office Tower	Reinforced Concrete	38	Coastal Miami	5	65.0	2,500 occupants
BL-002	Warehouse	Pre-Stressed Concrete	25	Inland Georgia	7	15.0	50 trucks/day
PP-001	Gas Turbine Plant	Ni-based Superalloy	30	Industrial NJ	5	120.0	500 MW
BR-004	Highway Bridge	Prestressed Concrete	50	Coastal Carolina	4	18.0	42,000 ADT
BR-005	Highway Bridge	A36 Steel + Concrete	55	Urban Northeast	3	22.0	55,000 ADT

Data Dictionary - Infrastructure Portfolio:

- **asset_id:** Unique identifier. Format: TYPE-NNN (BR = bridge, PL = pipeline, BL = building, PP = power plant).
- **type:** Infrastructure category. Determines which failure models and financial metrics apply.
- **material:** Primary structural material. Determines corrosion rate model, fatigue parameters (Paris' Law C and m), and failure mode (ductile vs. brittle).
- **age_years:** Years since construction. Primary driver of material degradation in the survival model.
- **location:** Geographic and environmental context. Determines corrosion environment: coastal = high chloride, industrial = chemical exposure, inland = moderate, urban = de-icing salt + pollution.
- **condition_rating:** Current condition on 1–9 scale (NBI standard). 9 = excellent, 7 = good, 5 = fair, 3 = serious, 1 = imminent failure. This is the target variable for condition prediction models.
- **loan_outstanding_M:** Outstanding loan balance in millions USD. Used for EAD computation. Amortisation schedule: 20-year term, level payments.
- **annual_traffic:** Usage intensity. ADT = Average Daily Traffic for bridges. BBL/day or BCF/day for pipelines. Occupants or trucks for buildings. Higher traffic = faster fatigue accumulation.

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DS-4: Sample Material-Financial Cross-Domain Features

The following dataset shows computed cross-domain features for 8 commodity-material pairs. These are the novel features that MatRisk AI generates at the intersection of material science and finance:

Commodity	MQI (0–100)	MQI 21D Trend	Supply Disruption Prob	Substitution Elasticity	Green Premium ($/kg)
Steel HRC	78.5	+2.3	0.18	0.42	+0.15
Aluminium LME	82.1	-0.8	0.12	0.55	+0.85
Copper COMEX	85.3	+0.5	0.15	0.28	+0.22
Nickel LME	71.2	-3.1	0.35	0.38	+1.20
Lithium	68.8	+1.7	0.42	0.22	+2.50
Cobalt	74.5	-1.2	0.55	0.65	+4.10
Rare Earth Index	62.3	-5.8	0.85	0.12	N/A
Iron Ore	80.0	+0.3	0.10	0.15	+0.08

Data Dictionary - Cross-Domain Features:

- **MQI:** Material Quality Index (0–100). Composite of CGNN-predicted quality metrics for the commodity's primary material products. 100 = highest quality. Computed daily from latest CGNN model outputs applied to production-representative compositions.
- **MQI_21D_trend:** 21-day rate of change in MQI. Positive = improving quality. Negative = degrading quality. This is the primary material science alpha signal for commodity prediction.
- **supply_disruption_prob:** Probability (0–1) of a significant supply disruption in the next 90 days. Based on: geographic concentration (Herfindahl index), production energy intensity, input material rarity, and geopolitical risk scores. Values > 0.5 indicate elevated risk.

- **substitution_elasticity:** Dimensionless (0–1). How easily this commodity's primary material can be replaced. 0 = no substitutes (rare earths for permanent magnets). 1 = readily substitutable (steel vs. aluminium for many structural applications). Computed from CGNN property similarity scores between the material and its closest alternatives.
- **green_premium:** Price differential ($/kg) between sustainably produced (recycled, low-carbon) and conventionally produced material. Positive = sustainable material costs more. Higher green premium = stronger ESG-driven demand signal for sustainable materials.

DS-5: Simulation Scenario Parameters

The following dataset defines the parameters for each of the 5 stress scenarios in the MatRisk Lab simulator and the stress testing engine:

Scenario	Shock Magnitude	Duration (quarters)	Affected Commodities	Affected Assets	Historical Precedent
Rare Earth Crisis	40% export reduction	3–4	Rare earths, lithium, cobalt	PP-001 (generators)	China 2010 export quota
Steel Quality Shift	15% tensile strength drop	2–3	Steel HRC, iron ore	BR-001, BR-002, BR-005	Quality issues at major mills
Climate Corrosion	30% corrosion rate increase	Permanent	All metals (indirect)	All coastal assets	IPCC +2°C projection
Recycled Mandate	50% recycled content required	4–6 (phase-in)	Aluminium, steel, copper	All (compliance cost)	EU Circular Economy Package
Novel Alloy Discovery	New alloy 20% stronger/cheaper	6–8 (adoption)	Steel, titanium	All steel bridges	Historical alloy transitions

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Using the Datasets

Interns must:

- Use DS-1 as a validation reference when downloading real Materials Project data - verify that API-returned values match these known values for common materials
- Use DS-2 format as the schema for all commodity data downloaded via yfinance. Extend to 10+ years and all 8 commodities.
- Use DS-3 as the exact portfolio for the MatRisk Lab simulator. These 10 assets appear in all game levels.
- Use DS-4 format as the output schema for the cross-domain feature engineering pipeline. Compute these features daily for each commodity.
- Use DS-5 as the parameter set for the stress testing engine. Each scenario must be reproducible from these parameters.
- All datasets must be version-controlled via DVC. Sample datasets go in data/external/. Real API data goes in data/raw/.

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SECTION A5: FEATURE ENGINEERING DEEP DIVE WITH WORKED EXAMPLES

A5.1 Magpie Compositional Descriptors - Worked Example

Matminer's Magpie featuriser extracts 132 statistical descriptors from a material's composition. For each elemental property (e.g., atomic radius, electronegativity), it computes six statistics across all elements in the composition weighted by their atomic fractions: mean, mean absolute deviation, minimum, maximum, range, and mode. Understanding these features is critical because they form the input to all tabular models and serve as global context for the GNN.

Worked Example - Titanium Alloy Ti-6al-4v:

Ti-6Al-4V is the most widely used titanium alloy in aerospace and medical implants. Its composition is approximately 90% Ti, 6% Al, 4% V by weight.

- **Step 1 - Convert to atomic fractions:** Molecular weights: Ti = 47.87, Al = 26.98, V = 50.94. Weight fractions: Ti = 0.90, Al = 0.06, V = 0.04. Atomic fractions: Ti = 0.90/47.87 / (0.90/47.87 + 0.06/26.98 + 0.04/50.94) = 0.8807, Al = 0.1040, V = 0.0153.
- **Step 2 - Compute Magpie features for electronegativity:** Pauling electronegativities: Ti = 1.54, Al = 1.61, V = 1.63. Weighted mean = 0.8807 × 1.54 + 0.1040 × 1.61 + 0.0153 × 1.63 = 1.549. This single feature captures the average electron-attracting tendency of the alloy. The maximum (1.63, V), minimum (1.54, Ti), and range (0.09) capture compositional spread.
- **Step 3 - Repeat for all 22 elemental properties:** Atomic number, atomic mass, atomic radius, boiling point, melting point, density, column, row, number of unfilled s/p/d/f orbitals, number of valence s/p/d/f electrons, electronegativity, specific heat, thermal conductivity, covalent radius, GSvolume_pa, GSbandgap, GSmagmom, SpaceGroupNumber. Total: 22 properties × 6 statistics = 132 features.

Financial Relevance Of Magpie Features:

Magpie features are not just abstract descriptors - they encode economically meaningful information. The mean atomic mass correlates with raw material cost (heavier elements tend to be rarer and more expensive). The electronegativity range indicates chemical reactivity (higher range → more reactive alloy → higher corrosion risk → shorter infrastructure RUL). The maximum melting point indicates high-temperature stability (relevant for power plant and aerospace insurance underwriting).

A5.2 Crystal Graph Construction - Worked Example

Worked Example - Sodium Chloride (NaCl, Rock Salt Structure):

NaCl crystallises in the cubic crystal system (space group Fm-3m, #225). The unit cell contains 4 Na atoms and 4 Cl atoms. Each Na atom is surrounded by 6 Cl atoms (and vice versa) at a distance of 2.82 Å.

- **Nodes:** 8 nodes total (4 Na + 4 Cl). Node features for Na: [atomic_number=11, electronegativity=0.93, radius=1.90Å, group=1, period=3, block=s]. Node features for Cl: [atomic_number=17, electronegativity=3.16, radius=0.99Å, group=17, period=3, block=p]. Feature dimension: 6 per node (before embedding).
- **Edges:** With 8Å cutoff radius, each atom has bonds to all atoms within this distance. Na-Cl bonds at 2.82Å (6 per atom), Na-Na and Cl-Cl second-nearest-neighbour bonds at 3.99Å (12 per atom). Edge features: Gaussian expansion of distance using 40 Gaussian basis functions centred at 0.5Å, 0.7Å, 0.9Å, ..., 8.3Å with width 0.5Å. This converts a single distance scalar into a 40-dimensional vector, enabling the GNN to learn distance-dependent interaction patterns.
- **Graph Statistics:** 8 nodes, ~144 directed edges (each atom has ~18 neighbours within 8Å, × 8 atoms). Data object size: approximately 0.5 KB per material. For 100K materials: ~50 MB total graph dataset.

A5.3 Financial Feature Engineering - Worked Example

Worked Example - Steel HRC Commodity Features:

Consider a 252-day window of Hot Rolled Coil (HRC) steel futures prices on CME. We compute the following feature set for the final day:

- **Returns:** Daily log return = ln(P_t / P_{t-1}). Weekly = sum of 5 daily returns. Monthly = sum of 21 daily returns. Annualised return = 252 × mean(daily returns).
- **Volatility (GARCH(1,1)):** Conditional variance: $\sigma^2_t = \omega + \alpha \cdot \varepsilon^2_{t-1} + \beta \cdot \sigma^2_{t-1}$. Estimate parameters (ω, α, β) via maximum likelihood on the 252-day window. Annualised volatility = $\sigma_t \times \sqrt{252}$. Typical HRC volatility: 25–40%.
- **Momentum (RSI-14):** RSI = 100 − 100/(1 + RS), where RS = average gain over 14 days / average loss over 14 days. RSI > 70: overbought (potential reversal). RSI < 30: oversold. For steel, RSI has historically been a weaker signal than for equities because commodity prices are driven more by supply fundamentals.
- **Mean Reversion (Bollinger Z-Score):** Z = (P_t − SMA_63) / σ_63. If Z > 2: price is 2 standard deviations above 63-day average (potential reversion). This feature captures the tendency of commodity prices to revert to production cost levels.

- **Term Structure:** Spread = F_2 − F_1 (second contract − front contract). Positive = contango (future price > spot, indicating ample supply/storage cost). Negative = backwardation (spot > future, indicating supply tightness). Steel often trades in backwardation during supply disruptions.

A5.4 Cross-Domain Fusion Features - Detailed Methodology

The cross-domain features represent MatRisk AI's core innovation - signals that exist only at the intersection of material science and finance:

Material Quality Index (MQI):

MQI is a composite score aggregating CGNN-predicted properties for materials in a specific commodity's production chain. For steel: MQI_steel = w_1 × (predicted tensile strength / benchmark) + w_2 × (predicted hardness / benchmark) + w_3 × (1 − predicted defect density / threshold). When a major steel producer's output MQI declines (detectable from material science signals before it appears in financial data), this predicts a quality premium compression in HRC futures.

Supply Disruption Probability:

Computed from material production difficulty metrics: energy required per kg (from CGNN thermodynamic predictions), rarity of constituent elements (from USGS reserve data), and geographic concentration of production (Herfindahl index of producing countries). For rare earth elements, the Supply Disruption Probability was at 0.85 in 2009 (pre-crisis) - correctly flagging extreme concentration risk. For aluminium, it's typically 0.15–0.25 (diverse global production base).

Substitution Elasticity:

Measures how easily a commodity's primary materials can be replaced by alternatives, based on CGNN-predicted property similarity scores between the target material and potential substitutes. Low substitution elasticity (e.g., rare earths for permanent magnets: elasticity = 0.1) means price is inelastic to demand shocks. High elasticity (e.g., steel vs. aluminium in automotive: elasticity = 0.6) means price shocks are dampened by substitution switching.

Green Premium Indicator:

The price differential between sustainably produced (recycled, low-carbon) and conventionally produced materials, normalised by the carbon intensity difference. Green Premium = (Price_recycled − Price_virgin) / (Carbon_virgin − Carbon_recycled). A rising green premium signals increasing ESG-driven demand for sustainable materials, creating trading opportunities in the recycled materials market.

SECTION A6: MODEL EVALUATION FRAMEWORK

A6.1 Material Property Prediction Metrics

All material property predictions are evaluated using the following metrics, consistent with published literature benchmarks:

Metric	Formula	Use Case	Target
Mean Absolute Error (MAE)	$\|y - \hat{y}\|$ averaged	Primary metric for all properties	Property-specific benchmarks
Root Mean Squared Error (RMSE)	$\sqrt{(\text{mean}((y - \hat{y})^2))}$	Penalises large errors more than MAE	$< 1.5 \times$ MAE
R^2 (Coefficient of Determination)	$1 - SS_{res} / SS_{tot}$	Overall model fit quality	> 0.90
Mean Absolute Percentage Error	$\|y - \hat{y}\| / \|y\|$ averaged	Scale-independent comparison	$< 10\%$
Physics Constraint Satisfaction	% predictions satisfying all constraints	Model physical plausibility	$> 95\%$

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A6.2 Financial Model Evaluation Metrics

Metric	Formula	Interpretation	Target
Sharpe Ratio	$(\mu - r_f) / \sigma$	Risk-adjusted return	> 0.5
Hit Rate	% of correct directional predictions	Forecast accuracy	> 52%
Maximum Drawdown	Max peak-to-trough decline	Worst-case loss	< 20%
Information Ratio	Active return / tracking error	Consistency of alpha	> 0.3
Profit Factor	Gross profit / gross loss	Win/loss magnitude ratio	> 1.2
Concordance Index	$P(risk_i > risk_j \mid T_i < T_j)$	Survival model discrimination	> 0.70
Brier Score	$Mean((p_i - o_i)^2)$	Probability calibration quality	< 0.15

A6.3 Worked Example - Backtesting The Commodity Prediction Model

Consider a walk-forward backtest of the steel HRC prediction model over 2 years (504 trading days):

- **Setup:** Training window: 252 days. Validation: 63 days. Test: 21 days. Step size: 21 days (monthly retraining). Total: 24 test periods × 21 days = 504 days.
- **Baseline (Traditional features only):** Sharpe = 0.32, Hit Rate = 51.2%, Max Drawdown = 24.5%. This represents the performance achievable with standard commodity technical analysis.
- **Enhanced (Traditional + Material Science features):** Sharpe = 0.67, Hit Rate = 54.8%, Max Drawdown = 18.1%. The material science signals (MQI trend, supply disruption probability, substitution elasticity) contribute an incremental Sharpe of 0.35.
- **Statistical Significance:** Paired t-test on monthly returns: t-statistic = 2.41, p-value = 0.024. The improvement is statistically significant at the 5% level. Bootstrap 95% CI for Sharpe difference: [0.12, 0.58].

- **Feature Importance:** Top 3 material science features by SHAP importance: (1) MQI 21-day trend ($\varphi = 0.15$), (2) Supply Disruption Probability ($\varphi = 0.11$), (3) Term Structure Slope × MQI interaction ($\varphi = 0.08$). The interaction feature is particularly notable - material quality signals are most predictive when the futures curve is in backwardation (supply tightness).

A6.4 Worked Example - Infrastructure Credit Risk Calculation

Consider a hypothetical infrastructure loan for a 45-year-old steel bridge in a coastal environment:

- **Bridge Characteristics:** Material: A36 structural steel. Age: 45 years. Location: Coastal Florida (high chloride exposure). ADT: 35,000 vehicles/day. Last inspection rating: 5/9 (fair condition). Loan outstanding: $12 million. Remaining loan tenor: 10 years.
- **Survival Model Inputs:** Steel tensile strength (predicted by CGNN): 400 MPa (original was 450 MPa - 10% degradation from corrosion). Corrosion rate: 0.08 mm/year (from environmental model). Remaining wall thickness: 12mm (original 18mm). Fatigue cycles: 45 years × 365 days × 35,000 = 575 million cycles.
- **Survival Curve Output:** $S(5|x) = 0.92$ (92% probability of surviving 5 more years). $S(10|x) = 0.78$ (78% probability of surviving the full loan tenor). $S(15|x) = 0.61$.
- **Financial Metrics:** PD at year 5 = 1 − 0.92 = 8.0%. PD at year 10 = 1 − 0.78 = 22.0%. LGD (ductile failure, repairable) = 0.35. EAD at year 5 = $8.4M (from amortisation). EL at year 5 = 0.08 × 0.35 × $8.4M = $235,200. Cumulative 10-year EL = $924,000 (summing annual EL estimates).
- **Comparison:** A traditional credit model using only financial ratios and industry default rates would estimate PD at ~3–4% for a performing infrastructure loan - significantly underestimating the material-science-informed PD of 8% at year 5. The difference ($235K vs ~$100K in annual EL) directly impacts loan provisioning and pricing.

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A6.5 Worked Example - Insurance Catastrophe Model

Consider an insurance portfolio of 100 coastal concrete buildings, each similar to Champlain Towers South:

- **Portfolio:** 100 buildings, average age 35 years, reinforced concrete construction, coastal Florida location. Average insured value: $50M per building. Total insured value: $5 billion.
- **Physics-Informed Failure Probability:** Using the survival model with corrosion rate inputs: average annual failure probability per building = 0.8% (derived from S(1|x) for each building). This compares to the actuarial estimate of 0.2% based on historical national building failure rates - the actuarial approach underestimates by 4x because it doesn't account for the specific coastal corrosion environment.
- **Loss Severity:** Log-normal distribution with $\mu = \ln(\$5M)$, $\sigma = 0.8$. Mean severity = $6.2M. 95th percentile severity = $18.7M.
- **Aggregate Annual Loss:** Compound Poisson: $\lambda = 100 \times 0.008 = 0.8$ expected failures/year. E[Aggregate Loss] = 0.8 × $6.2M = $4.96M/year.
- **TVaR at 99.5%:** Using 100,000 Monte Carlo simulations of the compound Poisson distribution: TVaR_{99.5%} = $42.3M. This is the reserve the insurer should hold against tail risk. The actuarial TVaR (using 0.2% failure rate) would be ~$11.5M - dangerously under-reserved by $30.8M.
- **Premium Implication:** Physics-informed annual premium per building: $49,600 (based on EL + risk loading). Actuarial premium: $12,400. The gap of $37,200 per building represents either mispriced risk (if the actuarial premium is charged) or competitive advantage (if the physics-informed premium is charged to correctly price the risk while competitors under-reserve).

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SECTION A7: ENSEMBLE METHODS AND TABULAR BASELINES

A7.1 XGBoost for Material Property Prediction

While Graph Neural Networks capture structural information, gradient-boosted decision trees remain extremely competitive for tabular material data (Matminer features). XGBoost and LightGBM often match or exceed GNN performance when the feature engineering is thorough - making them essential baselines and complementary models in the MatRisk AI ensemble.

Why Ensemble Methods Are Critical:

- Interpretability: Tree-based models produce feature importance rankings that material scientists and financial analysts can understand and validate. A SHAP value showing that "mean electronegativity" is the top predictor of corrosion resistance is actionable; a GNN attention weight on atom 47 is not.
- Speed: XGBoost inference is orders of magnitude faster than GNN inference (microseconds vs. milliseconds per prediction). For real-time commodity signal generation, ensemble models are essential.
- Robustness: Ensemble methods handle missing data natively, are robust to outliers, and require minimal hyperparameter tuning compared to deep learning models. They provide a reliable fallback when GNN training is unstable.
- Complementarity: Ensemble methods excel at tabular compositional features; GNNs excel at structural features. Combining both in a stacking ensemble captures both information sources.

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XGBoost Configuration for Material Properties

Hyperparameter	Default	Search Range	Rationale
n_estimators	500	100–2000	More trees for complex property relationships
max_depth	6	3–10	Deeper trees for nonlinear composition-property mappings
learning_rate	0.05	0.01–0.3	Lower rate + more trees = better generalisation
min_child_weight	3	1–10	Controls overfitting to rare compositions
subsample	0.8	0.5–1.0	Row sampling for variance reduction
colsample_bytree	0.8	0.3–1.0	Feature sampling prevents feature dominance
reg_alpha (L1)	0.1	0–10	Sparse feature selection
reg_lambda (L2)	1.0	0.1–100	Regularisation for correlated Magpie features

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A7.2 Stacking Ensemble Architecture

The final MatRisk AI material property prediction uses a stacking ensemble that combines the CGNN, PINN, XGBoost, and LightGBM predictions:

- **Level 0 (Base Models):** CGNN predicts properties from crystal graphs. PINN refines CGNN predictions with physics constraints. XGBoost predicts from Magpie compositional features. LightGBM predicts from Magpie features (different hyperparameters for diversity).
- **Level 1 (Meta-Learner):** A Ridge Regression model takes the 4 base model predictions as input and learns optimal weights for combining them. Trained on out-of-fold predictions from 5-fold CV to prevent information leakage.
- **Rationale:** The stacking ensemble outperforms individual models because CGNN captures structural patterns that tabular models miss, while XGBoost/LightGBM capture compositional patterns more efficiently. The PINN ensures physics compliance that pure data-driven models may violate. Ridge regression as the meta-learner is chosen for its simplicity and resistance to overfitting with only 4 input features.

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SECTION A8: MODEL EXPLAINABILITY AND REGULATORY COMPLIANCE

A8.1 Why Explainability Matters For Financial Applications

Financial regulators (Basel Committee, IAIS, SEC) increasingly require model explainability for any model used in credit decisions, insurance pricing, or trading strategies. A black-box model that predicts a bridge will fail in 5 years is useless to a loan officer unless they can explain WHY - which material properties are degraded, which environmental factors are dominant, and what the confidence level is. MatRisk AI implements comprehensive explainability at every layer.

A8.2 SHAP (SHapley Additive exPlanations)

SHAP values provide a unified framework for feature attribution based on cooperative game theory. For each prediction, every feature receives a SHAP value indicating its contribution to pushing the prediction away from the average.

Global Explanations:

Average |SHAP| across all predictions gives global feature importance. For the steel commodity prediction model, expected top features:

- MQI 21-day trend (material science): Most important alternative data feature. Rising MQI indicates improving steel quality, which predicts quality premium expansion.
- Supply Disruption Probability (material science): Second most important material feature. Elevated disruption probability predicts supply tightness and price increases.
- 63-day rolling volatility (traditional): Captures market regime and position sizing.
- Term structure slope (traditional): Contango/backwardation indicates supply-demand balance.
- MQI × Term structure interaction (cross-domain): Material quality signals are most predictive during backwardation (supply-tight markets).

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Local Explanations:

For a single prediction (e.g., “Predict steel HRC price will rise 2.3% over next 21 days”), the SHAP waterfall plot shows exactly which features pushed the prediction upward or downward. This is critical for trading desk compliance: a portfolio manager can see that the prediction is driven by (a) MQI improvement (+0.8% contribution), (b) supply disruption probability increase (+0.6%), (c) momentum (+0.5%), (d) mean reversion (−0.3%), (e) volatility regime (−0.3%). If the material science features are removed (ablation), the prediction drops to +0.9% - quantifying the material signal contribution for this specific trade.

A8.3 Partial Dependence Plots

Partial Dependence Plots (PDPs) show how a single feature affects predictions while marginalising over all other features. Key PDPs for MatRisk AI:

- **MQI vs. Commodity Return:** Expected to show a positive, roughly linear relationship. When MQI increases by 1 standard deviation, the predicted 21-day commodity return increases by approximately 0.3–0.5%.
- **Corrosion Rate vs. PD:** Expected to show an exponential relationship. Below 0.05 mm/year (mild corrosion), PD is relatively flat. Above 0.1 mm/year, PD increases sharply as the component approaches critical wall thickness loss.
- **Building Age vs. Insurance Loss Frequency:** Expected to show a bathtub curve: higher failure rate in early years (construction defects), lower in mid-life, and increasing in later years (material degradation).
- **Band Gap vs. Formation Energy:** A physics-validation PDP - should show that metals (band gap = 0) have lower formation energy (more stable), which validates that the model has learned fundamental physics.

A8.4 Counterfactual Explanations For Risk Decisions

Counterfactual explanations answer the question: "What is the minimum change to the input that would change the risk classification?" This is particularly important for credit decisions under regulatory scrutiny.

Worked Example - Bridge Credit Risk:

A bridge is currently classified as "investment grade" (PD < 5% at 5 years). The counterfactual explanation shows: "If the corrosion rate increased from 0.06 mm/year to 0.09 mm/year OR the average daily traffic increased from 25,000 to 42,000 vehicles, the bridge would be reclassified to 'speculative grade' (PD > 5%)." This gives the credit analyst two specific risk factors to monitor and thresholds to set for early warning triggers.

For insurance: "If the building's concrete cover thickness were reduced from 50mm to 35mm (simulating spalling damage), the annual failure probability would increase from 0.3% to 1.2%, requiring a premium increase from $18,000 to $54,000." This links material condition directly to insurance pricing in a way that is auditable and defensible to regulators.

PART B: GAMIFIED SIMULATION BUSINESS CONCEPT

Note on the Simulation Concept: The gamified simulation concept presented in this section is provided as a reference framework. Students may choose to follow this concept or design their own original gamified simulation - with a different name, structure, narrative, or game mechanics - as long as it meets the assessment criteria outlined in Part F.

SECTION B1: MATRISK LAB - SIMULATION OVERVIEW

B1.1 Business Rationale And Problem Statement

MatRisk Lab is a browser-based gamified simulation platform designed to train and certify material-financial analysts through realistic, scenario-driven challenges. The player assumes the role of a Quantitative Material-Financial Analyst at a fictional cross-domain intelligence firm - MatRisk Capital Partners - managing a portfolio of 10 infrastructure assets and 8 commodity trading positions. Each session simulates a full business quarter compressed into a 30-minute gameplay experience.

The Training Gap This Platform Solves:

There is currently no training platform that teaches the intersection of material science and financial analytics. Material science students learn crystallography and thermodynamics without understanding how their work affects commodity markets and infrastructure finance. Finance students learn portfolio theory and risk models without understanding the physical properties of the assets they price and insure. MatRisk Lab creates a zero-risk environment where learners experience material-financial decision-making and see the consequences of ignoring physical science in financial models.

Target Audience:

- **Primary:** Undergraduate and graduate students in engineering, data science, and finance preparing for quantitative roles at commodity trading desks, infrastructure investors, and insurers
- **Secondary:** Working professionals in commodity trading, project finance, and insurance seeking to integrate material science intelligence into their analytical frameworks
- **Tertiary:** Universities and corporate training programmes seeking to teach cross-domain analytical thinking in a gamified environment

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B1.2 Simulation Architecture

Core Components:

- **Infrastructure Portfolio:** Pre-built data for 10 fictional infrastructure assets - 3 steel bridges, 2 concrete bridges, 2 pipelines (one coastal, one inland), 2 commercial buildings, and 1 power plant - each with defined material composition, age, environmental exposure, and condition history
- **Commodity Book:** 8 commodity trading positions across steel (HRC), aluminium, copper, nickel, lithium, cobalt, rare earth index, and iron ore - with realistic position limits, margin requirements, and P&L tracking
- **Material Science Engine:** Embedded MatRisk AI models (CGNN, PINN, survival network) running in inference mode, providing real-time material property predictions and infrastructure RUL updates in response to scenario events
- **Market Simulator:** Dynamic commodity price generation using regime-switching models calibrated to historical volatility patterns, with material science events injecting exogenous shocks
- **Scenario Injector:** Randomised event engine that generates material-related financial shocks - supply disruptions, quality degradation events, regulatory changes, climate-accelerated corrosion, novel material discoveries
- **Insurance Module:** Players must maintain insurance coverage for their infrastructure portfolio, setting premiums and reserves based on their understanding of material failure risks
- **Scoring Engine:** Awards points for portfolio return, risk management quality (Sharpe ratio), insurance accuracy (loss ratio within target), and ESG compliance
- **Learning Layer:** Every decision triggers an explanation of the material science and financial concepts involved, with references to MatRisk AI model outputs

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B1.3 Game Modes

Campaign Mode (Primary Mode):

- 6 progressive levels, each spanning 4 simulated quarters (1 year of gameplay)
- Level 1: Single steel bridge + one commodity (steel HRC). Normal market. No material events. Learn basic portfolio monitoring and NAV computation.
- Level 2: Three infrastructure assets + three commodities. First material event: routine inspection reveals accelerated corrosion on the coastal pipeline. Player must assess financial impact.
- Level 3: Five assets + five commodities. Credit risk fund introduced. Rating downgrade scenario triggered by material failure at a major supplier. Player must hedge commodity exposure and adjust infrastructure risk scores.
- Level 4: Seven assets + all commodities. Rare earth supply crisis (China export restriction). Players must use material science intelligence (substitution analysis) to reposition commodity books while managing infrastructure assets affected by alternative material availability.
- Level 5: Full portfolio. Climate-accelerated corrosion scenario (+2°C projection). Multiple infrastructure assets see RUL shortening. Player must restructure insurance coverage, adjust credit provisions, and make strategic commodity trades based on material demand shifts.
- Level 6: Crisis mode - simultaneous material supply shock, credit event, and infrastructure emergency. The player competes against the AI opponent (using MatRisk AI models) for best portfolio performance under extreme stress.

Practice Mode:

- Free-play access to any single scenario type - player selects asset mix, scenario, difficulty, and duration
- Ideal for practising specific skills: commodity hedging with material signals, insurance pricing with physics-informed failure curves, ESG portfolio optimisation under material substitution constraints
- No penalties; unlimited attempts; step-by-step guidance and MatRisk AI model explanations available

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Assessment Mode:

- Timed, scored, and monitored session for formal evaluation during internship assessment or hiring
- No hints; no guidance; full scoring rubric applied; results compared against AI benchmark performance

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SECTION B2: SCORING SYSTEM AND GAME MECHANICS

B2.1 1000-Point Scoring Framework

Scoring Category 1 - Portfolio Return (250 Points):

- Full credit (250): Portfolio return exceeds the AI benchmark by >10%
- Partial credit (125–249): Portfolio return positive and within 10% of AI benchmark
- Minimum credit (50–124): Portfolio return positive but significantly underperforms AI
- Zero credit: Negative portfolio return

Scoring Category 2 - Risk Management (250 Points):

- Full credit (250): Sharpe ratio > 1.0 AND maximum drawdown < 15%
- Partial credit (125–249): Sharpe ratio > 0.5 OR maximum drawdown < 25%
- Zero credit: Sharpe ratio < 0 or drawdown > 40%

Scoring Category 3 - Material Science Application (250 Points):

- Full credit (250): Correctly identified all material-related financial risks before they materialised AND took appropriate actions (hedging, insurance adjustment, portfolio rebalancing)
- Partial credit (125–249): Identified most material risks but missed some or took suboptimal actions
- Zero credit: Failed to identify material risks or took counterproductive actions

Scoring Category 4 - Insurance & ESG (250 Points):

- Insurance accuracy (125): Loss ratio within 90–110% of expected. Premium pricing reflects material science risk assessment.
- ESG compliance (125): Portfolio ESG score improved over the game period AND material substitution decisions correctly balanced cost, properties, and environmental impact

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SECTION B3: SCENARIO ENGINE DESIGN

B3.1 Material-Financial Shock Scenarios

The scenario engine generates realistic material-related financial events based on historical precedents. Each scenario has a material science trigger, a financial impact chain, and a set of optimal player responses.

Scenario	Material Trigger	Financial Impact Chain	Optimal Response
Rare Earth Crisis	China restricts rare earth exports by 40%	EV material costs spike → battery metals → auto equity	Long rare earth, short lithium (substitution), adjust infrastructure assets using rare earth components
Steel Quality Shift	Major mill shifts to lower-grade output	Infrastructure repair acceleration → bridge PD increase → insurance claims spike	Increase insurance reserves, reduce infrastructure credit exposure, short HRC quality premium
Climate Corrosion	+2°C scenario accelerates corrosion 30%	Infrastructure RUL shortens → portfolio EL increase	Buy corrosion-resistant material futures, increase insurance premiums, provision for early maintenance
Recycled Mandate	EU mandates 50% recycled content	Virgin demand drops → price restructuring → stranded assets	Long recycled premium, short virgin metal, adjust ESG scores, review green bond eligibility
Novel Alloy Discovery	New alloy replaces steel in construction	Steel demand shock → substitution cascade	Analyse GAN inverse design output, adjust commodity book, re-estimate infrastructure RUL with new material

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B3.2 Scenario Difficulty Calibration

Each scenario has three difficulty levels that affect the complexity of the financial impact chain and the time available for response:

- **Standard:** Single material event with clear financial impact. 2 quarters to respond. Full model output available. Suitable for Level 1–3.
- **Advanced:** Multiple simultaneous material events with interacting financial impacts. 1 quarter to respond. Partial model output (player must request specific analyses). Suitable for Level 4–5.
- **Crisis:** Cascading material failures with systemic financial contagion. Immediate response required. Limited model availability (simulating real-world system overload). Suitable for Level 6 only.

B3.3 Worked Example - Rare Earth Crisis Scenario Gameplay

This walkthrough illustrates how a player would experience the Rare Earth Crisis scenario at Advanced difficulty (Level 4):

Quarter 1 (Setup):

- Player's portfolio: 7 infrastructure assets (3 bridges, 2 pipelines, 1 building, 1 power plant) and 7 commodity positions
- AI opponent holds the same initial portfolio but makes decisions using MatRisk AI models
- Market conditions: normal. All infrastructure assets in stable condition. Commodity book slightly profitable.

Quarter 2 (Trigger Event):

- News flash: China announces 40% reduction in rare earth export quotas, effective immediately
- Immediate market impact: neodymium price jumps 25% in 2 days (simulated from historical 2010 data)
- Player receives MatRisk AI alert: Supply Disruption Probability for rare earths has jumped from 0.45 to 0.92
- Player must decide: (a) go long rare earth index to profit from further price rise, (b) short downstream manufacturers (EV, wind turbine companies), (c) use GAN inverse

designer to identify substitute materials and long those commodity positions, (d) some combination

- AI opponent immediately runs GAN inverse design with elevated rare earth prices, identifies ferrite substitutes and iron nitride alternatives, goes long iron ore (ferrite input) and short rare earth (expecting eventual substitution dampening)

Quarter 3 (Cascade):

- Second-order effects materialise: infrastructure assets using rare-earth-containing components (power plant generators, wind turbines) face maintenance cost increases
- The power plant asset's RUL estimate shortens because replacement generators with rare earth magnets are now 3x more expensive, making the operator more likely to defer maintenance
- Player must adjust insurance reserves for the power plant and reassess the infrastructure credit risk exposure
- Commodity book: rare earths up 60% (from initial trigger), iron ore up 12% (substitution demand), lithium flat, cobalt down 8% (EV production slowdown)

Quarter 4 (Resolution):

- China partially reverses export restrictions under WTO pressure
- Rare earth prices decline 30% from peak but remain elevated (still 40% above pre-crisis)
- Player's portfolio performance vs. AI benchmark determines the score for this scenario
- Scoring: Did the player correctly identify the substitution opportunity? Did they adjust infrastructure risk exposures? Was the insurance reserve adjustment appropriate?

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SECTION B4: EDUCATIONAL LEARNING PATHS

B4.1 Learning Path 1 - Material Science Fundamentals

This learning path ensures players build a solid foundation in material science before encountering complex financial scenarios:

- **Module 1 - Crystal Structure Recognition:** Given a set of material properties, identify the crystal system and predict the space group. Score: accuracy of identification. Learning: relationship between structure and properties.
- **Module 2 - Property Prediction Challenge:** Given a composition (e.g., Ni_3Al), predict mechanical properties (bulk modulus, shear modulus) and compare against CGNN predictions. Score: closeness to model output. Learning: which compositional features drive which properties.
- **Module 3 - Failure Mode Analysis:** Given material properties and environmental conditions, predict the dominant failure mechanism (fatigue, corrosion, creep, brittle fracture) and estimate time-to-failure. Score: correct identification and RUL estimate accuracy.
- **Module 4 - Inverse Design Challenge:** Set target properties (tensile strength, density, cost) and use the GAN interface to generate candidate alloys. Evaluate generated compositions for physical plausibility. Score: how close generated compositions achieve targets while remaining within cost budget.

B4.2 Learning Path 2 - Financial Analytics

- **Module 5 - Commodity Signal Reading:** Interpret MatRisk AI commodity dashboard signals. Given a set of material science indicators (MQI trend, supply disruption probability, substitution elasticity), predict the commodity price direction. Score: directional accuracy.
- **Module 6 - Infrastructure Credit Scoring:** Given bridge inspection data, material composition, and environmental factors, compute PD/LGD/EL using the survival model outputs. Score: accuracy of financial metric computation.
- **Module 7 - Insurance Pricing Workshop:** Price an engineering liability insurance policy for a given structure using physics-informed failure probabilities. Compare against an actuarial benchmark. Score: premium accuracy and reserve adequacy.
- **Module 8 - ESG Portfolio Optimisation:** Given a portfolio of infrastructure assets, recommend material substitution changes that maximise ESG score improvement while minimising cost increase and property degradation. Score: ESG improvement per dollar of additional cost.

SECTION B5: LEADERBOARD AND CERTIFICATION

MatRisk Lab maintains a persistent leaderboard ranking players by cumulative score across all completed scenarios. Leaderboard categories:

- **Overall Rank:** Total score across all completed levels
- **Material Science Champion:** Highest score in material science application category (correctly identifying and responding to material-related risks)
- **Financial Analyst:** Highest Sharpe ratio achieved across all scenarios
- **Risk Manager:** Lowest maximum drawdown while maintaining positive returns
- **ESG Leader:** Greatest ESG score improvement across portfolio

Certification: Players who complete all 6 Campaign levels with a cumulative score > 4,000 (out of 6,000) earn the MatRisk Certified Analyst credential. This credential is verifiable through Zetheta's platform and indicates proficiency in material-financial cross-domain analytics.

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PART C: CASE STUDIES AND REAL-WORLD APPLICATIONS

SECTION C1: DEEPWATER HORIZON - MATERIAL FAILURE TO FINANCIAL CATASTROPHE (2010)

C1.1 Background

On 20 April 2010, the Deepwater Horizon drilling rig in the Gulf of Mexico experienced a catastrophic blowout that killed 11 workers, caused the largest marine oil spill in history (~4.9 million barrels), and resulted in over $65 billion in total financial costs for BP. The root cause was a material failure chain: the cement barrier (designed to seal the well) failed due to incorrect nitrogen foam composition, followed by failure of the blowout preventer's blind shear rams - which could not cut through the drill pipe because the pipe had buckled into a position that exceeded the rams' design shear capacity.

C1.2 Material Science Analysis

Cement Failure:

Halliburton's nitrogen foam cement was designed to create a lightweight barrier with sufficient compressive strength. However, testing showed the cement slurry was unstable at the wellbore's temperature and pressure conditions (246°F, 12,500+ psi). The material failure was predictable: the cement's compressive strength degraded under the specific thermodynamic conditions of the well. A PINN-based material model trained on cement behaviour under extreme conditions could have flagged this risk.

Blowout Preventer Failure:

The BOP's blind shear rams were designed to cut standard drill pipe under rated conditions. However, the pipe had buckled under pressure, presenting the rams with a cross-section that exceeded their rated shear capacity. The shear modulus of the drill pipe steel, combined with the buckled geometry, created a force requirement beyond the rams' hydraulic capacity. Material property prediction (shear modulus + failure mode analysis) could have identified this vulnerability.

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C1.3 Financial Impact Analysis

Direct Costs:

- Clean-up costs: $14 billion
- Government fines and penalties (Clean Water Act): $5.5 billion
- Settlements with businesses and individuals: $18.7 billion
- Securities litigation settlements: $7.1 billion
- BP share price decline: $105 billion in market capitalisation lost

Insurance Implications:

The Deepwater Horizon loss was one of the largest insured losses in history. Traditional insurance models based on historical drilling accident frequency massively underestimated the severity because they did not model the material failure chain (cement degradation → BOP shear capacity exceedance → uncontrolled blowout). A physics-informed catastrophe model that modelled cement behaviour under extreme conditions and BOP material capacity would have produced a significantly different (and more accurate) tail risk estimate.

C1.4 Matrisk AI Application

Students must analyse this case study through the MatRisk AI lens:

- Use the PINN module to model cement compressive strength degradation under Deepwater Horizon's wellbore conditions
- Use the survival model to estimate BOP failure probability given the known drill pipe material properties and the stress conditions
- Compute the insurance TVaR that a physics-informed model would have produced vs. the actuarial model that was actually used
- Estimate BP's credit risk (PD increase) that would have been triggered by a material-aware early warning system

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SECTION C2: MORANDI BRIDGE COLLAPSE - INFRASTRUCTURE CREDIT RISK FAILURE (2018)

C2.1 Background

On 14 August 2018, the Ponte Morandi (Polcevera Viaduct) in Genoa, Italy collapsed, killing 43 people. The bridge was a cable-stayed design completed in 1967, using a novel prestressed concrete structure designed by engineer Riccardo Morandi. The collapse was attributed to progressive corrosion-induced degradation of the stay cables' steel tendons, which had been losing cross-sectional area for decades due to environmental exposure (coastal Mediterranean climate, industrial pollution, de-icing salt usage).

C2.2 Material Science Analysis

Corrosion Mechanism:

The stay cables used high-strength steel tendons encased in prestressed concrete. Over 51 years of exposure, chloride ions from the marine atmosphere and de-icing salts penetrated the concrete cover and initiated pitting corrosion on the steel tendons. As corrosion reduced the steel cross-section, the remaining tendons experienced increasing stress - eventually exceeding their yield strength and triggering a brittle fracture cascade.

Predictability:

The corrosion rate of high-strength steel in a chloride-rich environment is well-characterised in material science literature. Given the bridge's material composition (steel grade, concrete cover thickness, chloride diffusion coefficient), environmental conditions (coastal exposure, temperature, humidity), and service history (traffic loading, maintenance records), a survival model trained on these features could have predicted the remaining useful life with actionable precision.

C2.3 Financial Impact Analysis

- Reconstruction cost: €202 million (new San Giorgio Bridge, completed 2020)
- Economic disruption: estimated €2–4 billion to Genoa's economy (major transport artery severed)
- Autostrade per l'Italia (operator) settlement with Italian government: €3.4 billion
- Insurance claims across property, business interruption, and liability lines
- Triggered European-wide infrastructure inspection mandates, increasing maintenance costs across the continent

C2.4 Matrisk AI Application

Students must demonstrate how MatRisk AI would have changed the financial outcome:

- Use the survival model with corrosion rate inputs to predict the bridge's RUL as of 2015 (3 years before collapse)
- Convert the survival curve to a PD curve and compute the expected loss for a hypothetical infrastructure loan on the Morandi Bridge
- Compare the physics-informed insurance premium (based on material degradation predictions) against the actuarial premium that was likely charged
- Model the financial impact on Autostrade's credit rating using the credit risk module

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SECTION C3: RARE EARTH SUPPLY CRISIS - COMMODITY MARKET DISRUPTION (2010–2011)

C3.1 Background

In September 2010, China - which controlled approximately 97% of global rare earth element production - restricted exports following a diplomatic dispute with Japan. Rare earth prices surged 10–20x within months: neodymium oxide went from $42/kg to $336/kg; dysprosium oxide went from $248/kg to over $2,800/kg. The crisis exposed the extreme material supply concentration risk in technology and clean energy sectors.

C3.2 Material Science Analysis

Substitution Analysis:

The crisis triggered intensive material science research into alternatives. For permanent magnets (the largest rare earth application): ferrite magnets could substitute but with 60–70% performance reduction; iron nitride emerged as a long-term alternative but required 5–10 years of development. For catalysts: cerium substitutes existed but with efficiency trade-offs. MatRisk AI's GAN inverse design module, with cost constraints reflecting the elevated prices, would have identified viable substitute compositions and quantified the property-cost trade-offs in real time.

C3.3 Financial Impact Analysis

- Rare earth-dependent company share prices dropped 15–40% during the initial crisis
- Companies with diversified material strategies (Toyota, with ferrite motor alternatives) recovered faster
- The crisis spawned a rare earth mining boom outside China (Lynas in Australia, Molycorp in USA) - but many projects failed when prices normalised
- Long-term impact: strategic stockpiling programmes established by Japan, EU, and USA

C3.4 Matrisk AI Application

- Use the commodity prediction model with material supply signals to forecast the price trajectory during the crisis
- Run the GAN inverse designer with elevated rare earth prices to generate substitute alloy compositions
- Use the substitution elasticity feature to identify which companies/sectors had the highest exposure
- Back-test the stress testing engine's "Rare Earth Crisis" scenario against actual 2010–2011 market data

SECTION C4: CHAMPLAIN TOWERS SOUTH - STRUCTURAL FAILURE AND INSURANCE (2021)

C4.1 Background

On 24 June 2021, Champlain Towers South, a 12-storey beachfront condominium in Surfside, Florida, partially collapsed, killing 98 people. The building, constructed in 1981, suffered from progressive deterioration of reinforced concrete elements, particularly in the pool deck area and parking garage. The concrete's reinforcing steel (rebar) had corroded significantly due to decades of exposure to salt-laden air and water intrusion.

C4.2 Material Science And Financial Analysis

The collapse shared the same fundamental material failure mechanism as the Morandi Bridge: chloride-induced corrosion of steel reinforcement in concrete. The corrosion reduced the rebar's cross-sectional area, eventually causing a punching shear failure in the pool deck slab that triggered progressive collapse of the residential floors above.

Financial Consequences:

- Insurance settlement: $1.02 billion (one of the largest residential structure settlements in US history)
- Triggered Florida SB 4-D: mandatory building recertification inspections at 25 years (previously 40 years) and every 10 years thereafter for buildings within 3 miles of the coast
- Insurance premium increases of 30–60% for aging coastal concrete buildings across Florida
- Estimated $1.5–3 billion in additional compliance costs for Florida building owners

C4.3 Matrisk AI Application

- Model the corrosion trajectory of the building's rebar using the survival network with environmental inputs (coastal Florida, 40-year exposure)
- Compute the physics-informed insurance premium that should have been charged vs. the actual premium
- Estimate the portfolio-level impact on Florida coastal building insurance using the catastrophe model with accelerated inspection data
- Design an early warning system: at what corrosion level would the survival model have triggered a financial risk escalation?

SECTION C5: NICKEL SHORT SQUEEZE - MATERIAL MARKET DISRUPTION (2022)

C5.1 Background

On 8 March 2022, nickel prices on the London Metal Exchange (LME) spiked over 250% in two days, reaching $101,365 per tonne - the largest commodity short squeeze in modern history. The trigger was geopolitical (Russia/Ukraine conflict and sanctions on Russian nickel), but the underlying dynamics were fundamentally about material science: the critical distinction between Class 1 nickel (high-purity, battery-grade) and Class 2 nickel (ferronickel, used in stainless steel) created artificial scarcity in the deliverable supply of Class 1 material.

C5.2 Material Science Analysis

Class 1 Vs Class 2 Nickel:

Class 1 nickel (>99.8% purity) is required for battery cathodes (NMC, NCA chemistries in EVs). Class 2 nickel (~20–40% Ni content in ferronickel) is used in stainless steel production. The LME's deliverable grade specification required Class 1 nickel - but global production had increasingly shifted toward Class 2 (Indonesia's HPAL processing). This meant the physically deliverable supply on LME was far smaller than total nickel production - a material-grade distinction that financial models overlooked.

Conversion Technology:

The possibility of converting Class 2 to Class 1 (via high-pressure acid leaching, HPAL, or matte-to-Class-1 refining) became the critical question. Material science determined the feasibility: HPAL produces mixed hydroxide precipitate (MHP) at ~40% Ni, which requires further refining. The conversion cost and timeline directly determined whether the short squeeze would persist or resolve.

C5.3 Financial Impact

- LME cancelled $3.9 billion in nickel trades - unprecedented in 145 years of operation
- Tsingshan Holding (Chinese nickel producer) faced $8+ billion in margin calls on short positions
- Nickel market illiquidity persisted for months; LME volumes dropped 70%
- Stainless steel producers and battery manufacturers faced extreme input cost volatility

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C5.4 Matrisk AI Application

- Use the Substitution Elasticity feature to quantify the real-time Class 1/Class 2 substitutability at different processing cost assumptions
- Model the GAN inverse designer with nickel-free battery cathode targets (LFP, LNMO) to estimate substitution switching timelines
- Use the Supply Disruption Probability feature to assess pre-crisis risk levels for nickel specifically
- Back-test: would the MQI and material supply signals have predicted the divergence between deliverable and total supply?
- Stress-test the portfolio under a nickel squeeze scenario to estimate the impact on infrastructure assets using nickel-containing stainless steel components

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PART D: PROJECT METHODOLOGY AND DELIVERABLES

SECTION D1: OVERVIEW

The project is structured as an AI-accelerated intensive sprint that compresses a comprehensive research and development initiative into a rigorous week-long programme. The intern will move from environment setup and data acquisition through model development, financial analytics integration, dashboard creation, and final documentation to produce a complete, production-quality GitHub repository.

The methodology uses AI-accelerated development - interns are encouraged to use Claude.ai, GitHub Copilot, and automated testing frameworks for code generation, debugging, and documentation. However, all model architectures must be understood and explainable by the intern, all physics constraints must be verified against primary sources, and all financial calculations must be validated against textbook formulas. Unverified AI output treated as final will be penalised.

Submission: Upon completion, the entire GitHub repository must be transferred to the ZethetaIntern organisation account via Settings → Danger Zone → Transfer Ownership. No personal forks or copies shall be retained post-transfer.

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SECTION D2: PART-BY-PART BREAKDOWN

Part 1 - Environment Setup & Data Acquisition:

Learning Goals: Establish the complete development environment, ingest data from all material science and financial sources, and produce initial exploratory analysis.

Tasks:

- Create the GitHub repository with the full directory structure (src/, data/, notebooks/, tests/, configs/, docker/, docs/)
- Set up CI/CD pipeline using GitHub Actions (lint with flake8/black, test with pytest on every push)
- Initialise DVC for data versioning and MLflow for experiment tracking
- Write data acquisition scripts for Materials Project API (100K+ entries), AFLOW API, and NIST datasets
- Acquire financial data: 10+ years of commodity prices (steel HRC, aluminium, copper, nickel, lithium, cobalt, rare earths, iron ore) via yfinance
- Download National Bridge Inventory (NBI) data - 620K+ bridge records with material type, condition ratings, inspection dates
- Produce 2 comprehensive EDA notebooks (one for material data, one for financial data) with 15+ publication-quality visualisations each
- Implement Great Expectations data validation suites for physical plausibility checking

Deliverable for Part 1: Complete GitHub repository with CI/CD, DVC-tracked data, 100K+ material entries, 10 years of commodity data, NBI bridge data, and 2 EDA notebooks.

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Part 1 Quality Gates:

(1) GitHub Actions CI pipeline must pass on push. (2) DVC status must show all data tracked. (3) MLflow UI accessible at localhost:5000. (4) EDA notebooks must render without errors. (5) Data quality report must document completeness > 80% for all critical fields.

Part 2 - Feature Engineering & Data Pipeline:

Learning Goals: Build the complete feature engineering pipeline for both material science and financial domains, including the novel cross-domain fusion features.

Tasks:

- Use Matminer to extract 132 Magpie compositional descriptors for all materials (atomic properties, electronic properties, thermodynamic properties)
- Build crystal structure → graph converter using Pymatgen: atoms as nodes (8–12 features), bonds as edges within 8Å cutoff (40–80 features via Gaussian expansion)
- Extract financial features: rolling returns, GARCH volatility, RSI, MACD, Bollinger bands, term structure slope for each commodity
- Build NBI decay feature extractor: per-bridge material-specific degradation curves from historical inspection records
- Create cross-domain fusion features: Material Quality Index (MQI), Supply Disruption Probability, Substitution Elasticity, Green Premium Indicator
- Implement point-in-time feature store with caching for expensive Matminer computations
- Run Great Expectations validation suites on all engineered features
- Achieve pytest coverage > 80% for all feature engineering modules

Deliverable for Part 2: Complete feature pipeline producing 132+ material features, 50+ financial features, crystal graph objects, and cross-domain fusion features. All validated and tested.

Part 2 Quality Gates:

(1) Matminer featurisation runs on a full dataset without errors. (2) Crystal graph objects load correctly in PyTorch Geometric DataLoader. (3) Financial features pass Great Expectations validation. (4) Cross-domain features show non-trivial correlation with commodity returns (preliminary check). (5) Feature store supports point-in-time queries without data leakage.

Critical Warning - Data Leakage Prevention:

Material science datasets contain families of similar compounds. If hematite appears in both training and test sets, the model will achieve artificially high accuracy because it memorises the iron oxide family's properties rather than learning generalisable composition-property relationships. Always split by chemical system (e.g., all Fe-O compounds in the same fold) to simulate predicting properties of truly novel materials. For financial data, never use future prices in feature computation (temporal leakage) - all features must be computable as-of the prediction date.

Part 3 - Core Model Development (GNN, PINN, ENSEMBLE):

Learning Goals: Implement and train the primary material property prediction models: Crystal Graph Neural Network, Physics-Informed Neural Network, and ensemble tabular models.

Tasks:

- Implement CGNN using PyTorch Geometric: 3-layer CGConv message passing, attention-based global pooling, multi-task prediction heads for band gap, formation energy, bulk modulus, and shear modulus
- Train CGNN with 5-fold cross-validation stratified by crystal system (NOT random splits - random splits cause massive data leakage in material science)
- Achieve benchmark performance: formation energy MAE $\leq$ 0.040 eV/atom, band gap MAE $\leq$ 0.40 eV, bulk modulus MAE $\leq$ 12 GPa, shear modulus MAE $\leq$ 10 GPa
- Implement PINN physics losses: elastic moduli consistency ($E = 2G(1+\nu) = 3K(1-2\nu)$), thermodynamic consistency ($G = H - TS$), Poisson's ratio bounds, Paris' Law for fatigue
- Joint-train CGNN + PINN with GradNorm for dynamic loss weight balancing
- Train XGBoost and LightGBM baselines on Matminer tabular features with Optuna HPO (50 trials each)
- Compute SHAP values for top 20 features per target property
- Log all experiments to MLflow with full reproducibility metadata (hyperparameters, metrics, artifacts)

Deliverable for Part 3: Trained CGNN + PINN meeting all benchmarks, trained ensemble baselines, SHAP analysis, comprehensive MLflow experiment log, model comparison notebook.

Part 3 Performance Benchmarks:

Property	Metric	Literature Benchmark	Our Target	Failure Threshold
Formation Energy	MAE (eV/atom)	0.033 (MEGNet, 2019)	≤ 0.040	> 0.060
Band Gap	MAE (eV)	0.33 (CGCNN, 2018)	≤ 0.40	> 0.60
Bulk Modulus	MAE (GPa)	10.5 (SchNet, 2018)	≤ 12.0	> 18.0
Shear Modulus	MAE (GPa)	8.2 (ALIGNN, 2021)	≤ 10.0	> 15.0
Physics Constraint Satisfaction	% predictions valid	Not reported	> 95%	< 85%

Note: If CGNN fails to meet benchmarks after 200 epochs, investigate: (1) graph construction - ensure cutoff radius is appropriate (8Å standard), (2) message passing depth - try 2-5 layers, (3) learning rate - try 1e-4 to 1e-2, (4) data quality - re-run Great Expectations to check for anomalous entries. Ensemble models (XGBoost/LightGBM) should serve as strong baselines - if they significantly outperform GNN, the graph construction may be suboptimal.

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Part 4 - Financial Analytics Modules:

Learning Goals: Build the complete financial analytics layer: commodity prediction with material signals, infrastructure credit risk, insurance catastrophe model, and ESG module.

Tasks:

- Build commodity price prediction model (LightGBM) with three feature groups: (A) traditional technicals, (B) material science signals from CGNN, (C) cross-domain interaction features
- Implement walk-forward backtesting: 252-day training window, 63-day validation, 21-day test, 10 bps transaction costs
- Conduct ablation study: baseline (A only) vs. enhanced (A+B+C) to quantify material signal alpha contribution
- Build Deep Survival Network for infrastructure RUL prediction on NBI bridge data (target: concordance index > 0.70)
- Implement PD/LGD/EL conversion module: survival curves → PD(t) = 1 − S(t|x), LGD from failure mode, EAD from loan amortisation
- Build insurance catastrophe model: failure probability from survival model × log-normal severity distribution = compound Poisson aggregate loss; compute TVaR at 99.5%
- Implement ESG module: carbon intensity calculator, substitution impact analyser, green bond eligibility scorer
- Write integration tests covering the full pipeline: raw data → features → model prediction → financial metrics

Deliverable for Part 4: Commodity prediction with demonstrated material signal alpha (Sharpe > 0.5), trained survival network, PD/LGD/EL module, insurance cat model, ESG module. All backtested and integration-tested.

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Part 4 Backtesting Parameters:

Parameter	Commodity Model	Survival Model	Insurance Model
Training Window	252 days (1 year)	All historical data pre-2018	10 years of claims data
Validation Window	63 days (1 quarter)	2018–2020 bridges	2 years held-out claims
Test Window	21 days (1 month)	2021–2023 bridges	Most recent year
Retraining Frequency	Monthly (21-day step)	Annual	Annual
Primary Metric	Sharpe ratio > 0.5	Concordance > 0.70	TVaR accuracy within 20%
Secondary Metric	Hit rate > 52%	Brier score < 0.15	Loss ratio 90–110%
Transaction Costs	10 bps round-trip	N/A	N/A

Ablation Study Protocol: To demonstrate that material science features genuinely improve commodity prediction, conduct a rigorous ablation study comparing four model variants: (A) Traditional features only, (B) Traditional + material science features, (C) Traditional + cross-domain interaction features, (D) Full model (A + B + C). Report Sharpe ratio, hit rate, and maximum drawdown for each variant. Use paired bootstrap test (10,000 resamples) to compute 95% confidence intervals for the Sharpe difference between (A) and (D). The ablation must show that material features provide statistically significant improvement - not just numerical improvement.

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Part 5 - GAN Inverse Design & Advanced Models:

Learning Goals: Build the GAN for inverse material design with financial cost constraints, portfolio stress testing engine, and supply chain risk module.

Tasks:

- Implement WGAN-GP with cost-constrained generator: noise + target properties + cost budget → composition vector via Softmax
- Train GAN with combined loss: Wasserstein loss + property matching loss (via frozen CGNN) + cost penalty (composition × element prices ≤ budget)
- Validate: generated compositions must be physically valid (positive fractions summing to 1), within cost budget 90%+ of the time, and achieve target properties within 10%
- Build Monte Carlo stress testing engine with 5 material-financial scenarios (rare earth crisis, steel quality shift, climate corrosion, recycled mandate, novel alloy disruption) - 10,000 simulation paths each
- Implement supply chain risk module: collateral quality scoring from material predictions, dynamic LTV haircut model, counterparty material risk assessment
- Run full Optuna HPO sprint: 100 trials per model for CGNN, PINN, GAN, survival network, and ensemble models
- Retrain all models with optimised hyperparameters and register final production models in MLflow model registry

Deliverable for Part 5: Trained WGAN-GP generating valid cost-constrained compositions, stress testing results for 5 scenarios, supply chain module, all models optimised and registered.

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Part 5 GAN Validation Criteria:

The GAN's output quality must be validated along three axes:

- **Physical Validity:** All composition fractions must be non-negative and sum to 1.0 (±0.001 tolerance). No element fraction should exceed 0.99 (pure element) unless explicitly requested. The CGNN-predicted properties of generated compositions must fall within physically plausible bounds (e.g., bulk modulus > 0, -1 < Poisson's ratio < 0.5). Target: >99% of generated compositions pass physical validity checks.
- **Property Achievement:** Generated compositions should achieve target properties within 10% tolerance for at least 80% of generation attempts. Use the frozen CGNN to evaluate properties of generated compositions. If property achievement is low, increase the property matching loss weight in the generator loss function.
- **Cost Compliance:** Generated compositions should fall within the specified cost budget for at least 90% of attempts. Cost = ∑(composition_fraction_i × current_element_price_i). Element prices sourced from USGS or commodity exchanges. If cost compliance is low but property achievement is high, the target property profile may be physically incompatible with the budget - flag this as a trade-off insight.

Stress Testing Methodology:

Each of the 5 stress scenarios follows a standardised simulation protocol:

- **Step 1 - Shock Calibration:** Define the material shock magnitude based on historical precedent (e.g., rare earth crisis 2010: 40% export reduction calibrated from actual Chinese quota announcements).
- **Step 2 - Financial Transmission:** Use the Material-Financial Signal Engine to translate the material shock into commodity price impacts, infrastructure RUL changes, and insurance loss adjustments.
- **Step 3 - Monte Carlo Simulation:** Run 10,000 paths sampling from the transmitted financial impact distributions with random noise. Each path produces a portfolio P&L outcome.
- **Step 4 - Risk Metrics:** Compute: expected P&L, VaR at 95% and 99%, TVaR at 99.5%, maximum loss, and recovery time distribution.
- **Step 5 - Sensitivity Analysis:** Vary the shock magnitude from 50% to 200% of the base case and plot risk metrics as a function of shock severity to identify non-linear breakpoints.

Part 6 - Dashboard, Simulator & Explainability:

Learning Goals: Build the interactive MatRisk AI dashboard, gamified simulation prototype, and comprehensive model explainability suite.

Tasks:

- Build 5-page Streamlit dashboard: (1) Material Property Explorer with SHAP explanations, (2) Commodity Signal Dashboard with backtest overlay, (3) Infrastructure Risk Scorer with survival curves, (4) Inverse Material Designer with Pareto frontier, (5) ESG Impact Analyser with substitution scenarios
- Build MatRisk Lab gamified simulator prototype: at minimum, implement Campaign Mode Level 1–2 with single infrastructure asset + commodity position, scenario injection, and basic scoring
- Implement SHAP explanations (global + local) for all models: TreeExplainer for ensemble, DeepExplainer for neural networks
- Generate partial dependence plots for top 10 features per model
- Run physics constraint audit: report percentage of predictions satisfying all physics constraints (target: > 95%)
- Produce comprehensive model performance report: CV results with confidence intervals, holdout test, financial backtests, ablation study, computational cost analysis

Deliverable for Part 6: Functional 5-page dashboard, MatRisk Lab simulator prototype (Levels 1–2), SHAP + PDP analysis, physics audit, performance report.

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Part 6 Dashboard Specification:

Each dashboard page must include the following minimum elements:

- **Material Property Explorer:** Chemical formula input field, crystal structure 3D viewer (if feasible, otherwise 2D lattice diagram), CGNN multi-property prediction with uncertainty bars, SHAP waterfall plot for each prediction, comparison table against 5 most similar materials from the database. Interactive: user can modify composition and see predictions update in real time.
- **Commodity Signal Dashboard:** Candlestick price chart with technical indicators overlay, MatRisk material signal panel (MQI trend, Supply Disruption Probability gauge, Substitution Elasticity bar), 21-day forward prediction with 90% confidence band, cumulative backtest P&L chart, feature importance bar chart with toggle for material vs. traditional features.
- **Infrastructure Risk Scorer:** Portfolio upload (CSV) or dropdown selector for NBI bridges, per-asset survival curve with 95% CI shading, PD/LGD/EL table with colour-coding (green/yellow/red thresholds), portfolio aggregate risk summary, stress test scenario selector with impact waterfall chart.
- **Inverse Material Designer:** Target property sliders (tensile strength, density, cost budget), "Generate" button triggering GAN inference, results table of top 5 candidate compositions with predicted properties and costs, Pareto frontier scatter plot (property score vs. cost), physics validation indicators (green check / red X).
- **ESG Impact Analyser:** Material substitution scenario builder (dropdown for original → substitute), carbon intensity comparison bar chart, property impact table (before vs. after), cost delta indicator, portfolio-level ESG score gauge (before vs. after), green bond eligibility checklist with threshold indicators.

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Part 7 - Documentation, Testing & Github Transfer:

Learning Goals: Write comprehensive documentation, achieve test coverage targets, Dockerise the application, and execute the GitHub repository transfer.

Tasks:

- Write comprehensive README.md: project overview, architecture diagram, quick start guide, data acquisition instructions, model training pipeline, dashboard launch, results summary, contributing guidelines, IP attribution
- Generate Sphinx API documentation: every public class/function documented with Google-style docstrings, type hints, and usage examples
- Write 15–20 page technical report (Markdown or LaTeX): problem formulation, literature review, model architectures with mathematical formulations, experimental results with statistical significance, limitations, future work, and 20+ references
- Achieve pytest coverage > 80% across all modules: unit tests, integration tests, data validation tests, model regression tests, property-based tests (Hypothesis library)
- Create Docker configuration: docker-compose.yml with API service (FastAPI), dashboard service (Streamlit), and MLflow service
- Pre-transfer checklist: all code committed, all tests passing, DVC data pushed, no personal API keys in repo, LICENSE file present, .gitignore configured
- Execute transfer: final commit with message “Final submission - MatRisk AI v1.0 - [Name] - [Date]”, create git tag v1.0, transfer ownership to ZethetaIntern via Settings → Danger Zone → Transfer Ownership
- Send confirmation email to project supervisor with transfer timestamp and repository URL

Deliverable for Part 7: Complete documentation, >80% test coverage, Docker configuration, and successfully transferred repository under github.com/ZethetaIntern/matrisk-ai.

Part 7 Technical Report Outline:

The 15–20 page technical report must follow this structure:

- **Chapter 1 - Introduction (2 pages):** Problem statement, convergence thesis (material science × finance), research questions, and contribution summary.
- **Chapter 2 - Literature Review (3 pages):** Prior work in (a) ML for material property prediction (CGCNN, MEGNet, ALIGNN, SchNet), (b) physics-informed ML (PINNs, physics-constrained optimisation), (c) alternative data in finance (satellite imagery, NLP for earnings calls), (d) infrastructure risk modelling. Minimum 20 references.
- **Chapter 3 - Methodology (4 pages):** Data sources and preprocessing, feature engineering pipeline, model architectures with mathematical formulations (CGNN message passing equations, PINN loss functions, GAN objective, survival model hazard function), training procedures, and evaluation protocols.
- **Chapter 4 - Results (4 pages):** Material property prediction benchmarks with tables and plots, financial backtesting results with equity curves, ablation study with statistical significance tests, physics compliance audit results, dashboard screenshots.
- **Chapter 5 - Discussion (2 pages):** Interpretation of results, comparison with baselines, practical implications for financial institutions, limitations (data quality, model assumptions, computational constraints), and potential failure modes.
- **Chapter 6 - Conclusion and Future Work (1 page):** Summary of contributions, immediate next steps (patent filing, client pilots), and long-term research directions (multi-modal material data, real-time sensor integration, production deployment).

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Part 7 Github Transfer Step-By-Step:

1. **Final Commit:** Stage all files, commit with message: "Final submission –Composit-MatRisk AI v1.0 - [Team Full Name] - [YYYY-MM-DD]". Push to main branch.
2. **Create Release Tag:** git tag -a v1.0 -m "MatRisk AI v1.0 Final Submission" && git push origin v1.0
3. **Verify CI/CD:** Confirm GitHub Actions shows green checkmark on the final commit. If any checks fail, fix and recommit before proceeding.
4. **Transfer Ownership:** Click on the task pane of www.zetheta.com – there is a button on transfer repository or Go to repository Settings → scroll to Danger Zone → click Transfer ownership → type repository name to confirm → enter new owner: ZethetaIntern → confirm transfer.
5. **Verify Transfer:** Navigate to github.com/ZethetaIntern/matrisk-ai and confirm the repository appears with all commits, branches, tags, and CI/CD configuration intact.
6. **Clean Local:** Delete ALL local clones, forks, and copies of the repository from personal devices. Remove any cached credentials or SSH keys associated with the repository.
7. **Confirmation:** Send email to project supervisor with: (a) transfer timestamp, (b) repository URL, (c) git tag v1.0 hash, (d) pytest coverage report summary, (e) brief self-assessment of work quality.

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SECTION D3: MANDATORY DELIVERABLES CHECKLIST

#	Deliverable	Acceptance Criteria
1	GitHub repository with full structure	All directories, CI/CD, DVC, MLflow
2	Material data ingestion (100K+ entries)	From Materials Project + AFLOW APIs
3	Financial data ingestion (10+ years)	Commodity prices + NBI bridge data
4	EDA notebooks (2)	30+ visualisations total
5	Matminer feature pipeline	132+ features per material, tested
6	Crystal graph builder	Converts structures to PyG Data objects
7	Financial feature pipeline	50+ features across price, volatility, momentum
8	Cross-domain fusion features	MQI, substitution elasticity, green premium
9	Trained CGNN meeting benchmarks	MAE targets met on all 4 properties
10	PINN physics constraints	All 4 constraints implemented and reducing violations
11	Ensemble baselines (XGBoost/LightGBM)	Trained, tuned, SHAP computed
12	Commodity prediction with ablation	Material signal alpha quantified, Sharpe > 0.5
13	Deep Survival Network	Concordance index > 0.70 on NBI data
14	PD/LGD/EL conversion module	Produces financial risk metrics from survival curves
15	Insurance catastrophe model	Aggregate loss distribution, TVaR at 99.5%
16	ESG module	Carbon intensity + green bond scoring

17	WGAN-GP with cost constraints	Valid compositions within budget 90%+ of time
18	Stress testing engine (5 scenarios)	Monte Carlo with 10K paths each
19	5-page Streamlit dashboard	All pages functional with real model outputs
20	MatRisk Lab simulator prototype	Campaign Levels 1–2 playable with scoring
21	SHAP + PDP explainability suite	Global + local for all models
22	Comprehensive README.md	All sections present, architecture diagram
23	Sphinx API documentation	100% public API documented
24	Technical report (15–20 pages)	Literature review, formulations, results, 20+ refs
25	Test coverage > 80%	pytest-cov report generated
26	Docker configuration	docker-compose up runs all services
27	Repository transferred to ZethetaIntern	Confirmed under ZethetaIntern org

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PART E: TOOLS AND PLATFORMS GUIDE

SECTION E1: DATA SCIENCE AND ML TOOLS

E1.1 Python Environment

- **Python 3.11+:** Primary language for all development. Use virtual environments (venv or conda) for dependency isolation.
- **PyTorch 2.x:** Deep learning framework for CGNN, PINN, GAN, and DeepSurv models. GPU acceleration via CUDA for training.
- **PyTorch Geometric:** Extension library for graph neural networks. Provides CGConv, message passing layers, and graph data structures.
- **XGBoost / LightGBM:** Gradient-boosted decision tree libraries for tabular feature prediction. LightGBM preferred for speed; XGBoost for comparison.
- **Scikit-learn:** Standard ML library for preprocessing, cross-validation, and baseline models.
- **Optuna:** Hyperparameter optimisation framework with Bayesian optimisation and pruning for efficient search.

E1.2 Material Science Libraries

- **Pymatgen:** Python Materials Genomics library. Crystal structure analysis, space group identification, Pourbaix diagrams, and Materials Project API client.
- **Matminer:** Automated feature extraction for materials. Provides Magpie, JARVIS, and Orbital descriptors. Can extract 132+ compositional features automatically.
- **ASE (Atomic Simulation Environment):** For manipulating atomic structures, reading/writing crystal structure files (CIF, VASP, etc.), and basic property calculations.

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E1.3 Financial Analytics Libraries

- **yfinance:** Yahoo Finance API wrapper for commodity price data. Supports OHLCV, futures, and options data.
- **lifelines:** Survival analysis library. Cox Proportional Hazards, Kaplan-Meier, Weibull models. Concordance index computation.
- **SHAP:** SHapley Additive exPlanations for model interpretability. TreeExplainer (fast, for tree models), DeepExplainer (for neural networks).
- **Statsmodels:** Statistical modelling. GARCH volatility models, time series analysis, and regression diagnostics.

E1.4 Mlops And Devops Tools

- **MLflow:** Experiment tracking, model registry, and model serving. Log hyperparameters, metrics, and artifacts for every training run.
- **DVC (Data Version Control):** Git for data. Version-control large datasets and model files. Ensures full reproducibility of data pipelines.
- **Docker:** Containerisation for reproducible deployment. Create Dockerfiles for API, dashboard, and MLflow services.
- **GitHub Actions:** CI/CD pipeline. Run lint (flake8, black), type checking (mypy), and tests (pytest) on every push.
- **pre-commit:** Git hook framework. Run formatters and linters automatically before every commit.

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SECTION E2: DETAILED TOOL TUTORIALS

E2.1 Pymatgen - Crystal Structure Analysis Tutorial

Pymatgen is the essential library for working with crystal structures in MatRisk AI. Key operations:

Loading Structures From Materials Project:

- Install: pip install mp-api pymatgen matminer
- API key: Register at materialsproject.org, obtain API key, set environment variable MP_API_KEY
- Basic query: mpr = MPRester(api_key); docs = mpr.summary.search(formula="Fe2O3", fields=["structure", "band_gap", "formation_energy_per_atom"])
- Bulk download: For large datasets, use pagination with chunk_size=1000 and checkpoint resumption to handle API rate limits

Structure Analysis:

- Space group: structure.get_space_group_info() returns (symbol, number) e.g., ('Fm-3m', 225) for NaCl
- Crystal system: structure.get_crystal_system() returns one of 7 systems (cubic, hexagonal, etc.)
- Lattice parameters: structure.lattice.abc returns (a, b, c) in Å; structure.lattice.angles returns (α, β, γ) in degrees
- Nearest neighbours: from pymatgen.analysis.local_env import VoronoiNN; nn = VoronoiNN(); neighbours = nn.get_nn_info(structure, 0) returns neighbour list for atom 0
- Bond analysis: from pymatgen.analysis.graphs import StructureGraph; sg = StructureGraph.with_local_env_strategy(structure, VoronoiNN()) creates the crystal graph directly

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E2.2 Matminer - Feature Extraction Tutorial

Magpie Featuriser:

- from matminer.featurizers.composition import ElementProperty
- featurizer = ElementProperty.from_preset('magpie')
- df = featurizer.featurize_dataframe(df, col_id='composition') - adds 132 columns automatically
- Processing time: approximately 1–2 minutes per 10,000 materials on a modern laptop. For 100K+ materials, use parallel processing: featurizer.set_n_jobs(4)
- Missing values: some elemental properties are undefined for certain elements (e.g., no electronegativity for noble gases). Matminer handles these gracefully with NaN - but you must implement imputation (median per crystal system) before training

Structural Featuriser:

- from matminer.featurizers.structure import DensityFeatures, GlobalSymmetryFeatures, SiteStatsFingerprint
- DensityFeatures(): adds density, volume per atom, packing fraction
- GlobalSymmetryFeatures(): adds space group number, crystal system, is_centrosymmetric
- SiteStatsFingerprint: computes statistics over per-site descriptors (coordination number, local order parameters)

E2.3 Pytorch Geometric - Graph Data Tutorial

Converting Crystal Structures To Pytorch Geometric Data Objects:

- Each material becomes a torch_geometric.data.Data object with: x (node features, shape [n_atoms, node_dim]), edge_index (connectivity, shape [2, n_edges]), edge_attr (edge features, shape [n_edges, edge_dim]), y (target properties, shape [n_targets])
- Node features: one-hot atomic number (up to 100 elements) + continuous features (electronegativity, radius, etc.). Embed one-hot via nn.Embedding for efficient representation.
- Edge construction: for each atom, find all atoms within cutoff radius (8Å). Use Pymatgen's structure.get_neighbors(site, r=8.0). Create bidirectional edges (i→j and j→i).

- Edge features: Gaussian expansion of distance: exp(-((d - μ_k) / σ)^2) for k = 1..40 with μ_k evenly spaced from 0.5Å to 8Å and σ = 0.5Å. This converts distance from scalar to rich 40-dimensional representation.
- Batching: torch_geometric.data.DataLoader handles variable-size graphs automatically by offsetting edge indices and concatenating features. Batch size 32–64 is typical for crystal graphs.

E2.4 MLflow - Experiment Tracking Tutorial

Setting Up Mlflow:

- Start server: mlflow server --backend-store-uri sqlite:///mlflow.db --default-artifact-root ./mlruns --host 0.0.0.0 --port 5000
- Set tracking URI: mlflow.set_tracking_uri('http://localhost:5000')
- Create experiment: mlflow.set_experiment('matrisk-cgnn')

Logging A Training Run:

- with mlflow.start_run(run_name='cgnn-v1-lr1e3'): start a tracked run
- mlflow.log_params({"lr": 1e-3, "hidden_dim": 128, "n_layers": 3}) - log hyperparameters
- mlflow.log_metric("val_mae_bandgap", 0.38) - log validation metric
- mlflow.log_artifact("shap_plot.png") - log visualisation artifact
- mlflow.pytorch.log_model(model, "model") - log the trained model for reproducibility
- Every training run must be logged. At project end, the MLflow UI should show a complete history of all experiments with comparable metrics across model versions.

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SECTION E3: DESIGN AND DOCUMENTATION TOOLS

- **Streamlit:** Dashboard framework for interactive web applications in pure Python. Key features: st.plotly_chart for interactive Plotly figures, st.slider for user inputs, st.dataframe for tabular display, st.selectbox for dropdown menus. Multi-page apps use st.set_page_config and pages/ directory structure.
- **Plotly:** Interactive visualisation library. 3D scatter (crystal structures), candlestick charts (commodities), heatmaps (correlations), survival curves with CI shading. Use Plotly Express for quick plots; Graph Objects for customisation.
- **Sphinx:** Documentation generator. Autodoc extension auto-generates from docstrings. Configure with sphinx-quickstart. Build: sphinx-build -b html docs/ docs/_build/.
- **Mermaid:** Diagram-as-code tool. GitHub renders natively in markdown. Architecture diagrams, data flow charts, state machines for game logic.
- **LaTeX / Overleaf:** For technical report if high-quality mathematical typesetting needed. Alternative: Markdown with MathJax rendering.

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SECTION E4: REPOSITORY STRUCTURE SPECIFICATION

The repository must follow this exact structure. Each directory has a specific purpose:

- matrisk-ai/ - Root. Contains README.md, setup.py, requirements.txt, Makefile, .pre-commit-config.yaml, .gitignore, LICENSE
- .github/workflows/ - CI/CD: ci.yml (lint + test on push), cd.yml (Docker build on tag)
- data/ - Subdirectories: raw/ (DVC-tracked), interim/ (cleaned), processed/ (model-ready), external/ (reference). NEVER commit raw data to git.
- notebooks/ - Subdirectories: 01_eda/, 02_feature_eng/, 03_modelling/, 04_evaluation/. All must render without errors.
- src/data/ - Data loaders: materials_project.py, aflow_loader.py, nbi_loader.py, commodity_loader.py
- src/features/ - Feature engineering: magpie_features.py, crystal_graph.py, financial_features.py, fusion_features.py, feature_store.py
- src/models/cgnn/ - Crystal Graph Neural Network: model.py, train.py, predict.py
- src/models/pinn/ - Physics-Informed NN: physics_losses.py, model.py, gradnorm.py
- src/models/gan/ - Inverse Designer: generator.py, critic.py, inverse_designer.py, cost_constraint.py
- src/models/survival/ - Deep Survival: deep_surv.py, cox_ph.py, financial_conversion.py
- src/models/ensemble/ - Tabular models: train_xgb.py, train_lgbm.py, stacking.py
- src/financial/ - commodity_predictor.py, credit_risk.py, cat_model.py, esg_module.py, stress_testing.py
- src/dashboard/ - Streamlit: app.py, pages/ (one file per dashboard page)
- src/simulator/ - MatRisk Lab: game_engine.py, scenario_injector.py, scoring.py, ai_opponent.py
- tests/ - Mirrors src/ structure. conftest.py for shared fixtures. Minimum coverage: 80%.
- configs/ - YAML hyperparameter configs. All hyperparameters in config files, never hardcoded.
- docker/ - Dockerfile.api, Dockerfile.dashboard, docker-compose.yml
- docs/ - Sphinx source. conf.py, index.rst, api/ (auto-generated)

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Makefile Commands (Mandatory):

- make install - Create venv, install all dependencies
- make lint - Run flake8, black --check, mypy
- make test - Run pytest with coverage report
- make data - Run all data acquisition scripts
- make features - Run feature engineering pipeline
- make train - Train all models with current configs
- make dashboard - Launch Streamlit at localhost:8501
- make docker - Build and start all Docker services
- make docs - Generate Sphinx documentation
- make all - Full pipeline: install → data → features → train → dashboard

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SECTION E5: RESEARCH AND DATA SOURCES

E5.1 Material Science Research

- **Materials Project Website:** materialsproject.org - browse materials, download data, and access the API documentation
- **AFLOW:** aflow.org - largest computational materials database with REST API
- **NIST:** nist.gov/srd - Standard Reference Data for experimentally validated material properties
- **arXiv (cond-mat):** arxiv.org/list/cond-mat - latest research papers on computational materials science

E5.2 Financial Data Sources

- **National Bridge Inventory:** fhwa.dot.gov/bridge/nbi - 620K+ US bridge records with material type and condition data
- **USGS Mineral Commodities:** usgs.gov/centers/national-minerals-information-center - production, price, and reserve data for 90+ minerals
- **World Bank Open Data:** data.worldbank.org - ESG indicators, carbon intensity, and sustainability metrics

E5.3 AI Research Tools

- **Claude.ai:** For research synthesis, code generation, debugging, and documentation drafting. All outputs must be verified against primary sources.
- **GitHub Copilot:** AI-assisted code completion. Accelerates boilerplate code writing but requires review for correctness.
- **Perplexity.ai:** For quick factual research and finding specific papers, datasets, and API documentation.

Critical: AI tools are accelerators, not replacements for understanding. Every model architecture, physics constraint, and financial formula used in the project must be explainable by the intern without AI assistance. Unverified AI-generated content treated as final work product will result in point deduction.

PART F: SUBMISSION PROTOCOL

Projects are assessed using a comprehensive 1000-point rubric covering seven dimensions: Problem Understanding, Solution Quality, Research & Analysis, Presentation & Clarity, Innovation & Creativity, Feasibility & Practicality and CV alignment. Assessment is conducted through a combination of manual review and AI-powered evaluation.

SECTION F1: AUTOMATIC DISQUALIFIERS

The following issues result in automatic project rejection requiring re-submission:

- Repository not transferred to ZethetaIntern by the deadline
- Personal API keys, passwords, or credentials committed to the repository
- Test coverage below 60% (hard minimum)
- Plagiarism: code copied from existing repositories without attribution and modification
- Data leakage: training data contaminating the test set (especially random splits on material data)
- Non-functional Docker build: docker-compose up must start all services without errors
- Missing technical report or README
- Personal forks or copies retained after transfer (verified by Zetheta's security team)

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SECTION F2: SUBMISSION FINAL CHECKLIST

Before executing the repository transfer, verify every item:

- All code committed and pushed to main branch (no dangling branches with unmerged work)
- All tests passing in CI/CD pipeline (green checkmark on latest commit)
- DVC remote configured and all data files pushed
- MLflow experiments exported and committed
- Docker build succeeds without errors (test with docker-compose up --build)
- README.md is complete with architecture diagram and screenshots
- Technical report is complete (15–20 pages, 20+ references)
- Sphinx API documentation generated and committed to docs/
- No personal API keys or credentials in repository (use .env.example template)
- LICENSE file present with Zetheta IP attribution
- .gitignore properly configured (no data files, no __pycache__, no .env, no mlruns/)
- Final commit message: "Final submission – MatRisk AI v1.0 — [Your Email Id] — [Date]"
- Git tag created: git tag -a v1.0 -m "MatRisk AI v1.0 Final Submission"
- Repository to be transferred through Zetheta task pane in the projects ideally – if not possible then Repository transferred via Settings → Danger Zone → Transfer Ownership → ZethetaIntern
- Confirmed repository appears under github.com/ZethetaIntern/matrisk-ai
- All local forks, copies, and clones deleted from personal devices
- Confirmation email sent to project supervisor with transfer timestamp and URL

Post-Transfer: ZethetaIntern will review the repository and score/ rank the submissions.

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IP PROTECTION PROTOCOL:

All IP arising from this project including submissions made is the exclusive property of Zetheta Algorithms Pvt. Ltd. as per the confidentiality terms accepted at project commencement

- All innovations documented with dated lab notebook entries in the technical report and git commit history
- Novel algorithms include pseudo-code and mathematical formulation in patent-ready format
- Comparative results against prior art quantified and documented in the model performance report
- The GitHub repository commit history serves as a timestamped invention record under Indian patent law

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END OF PROJECT DOCUMENT